

Executive Summary

Funding for large-scale infrastructure projects often involves partnerships from public entities at the local, regional, and/or national levels. However, unforeseen challenges thwart full-fledged financing of projects that use public funds. This is because public funds are often politically invested, temporal, tax-dependent, and sensitive to the public agency's economic health. Large-scale projects like high-speed rail (HSR) require funding from a variety of avenues beyond those available through public monies. Although HSR serves the general public's mobility needs, any funds (whether State or Federal) flowing from the public exchequer usually undergo strict review and scrutiny. Funds from public agencies are always limited, making such traditional financing mechanisms unsustainable for fulfilling HSR's long-term operational and maintenance cost needs—on top of initial costs involved in construction. Therefore, any sustainable means of financing HSR projects would always be welcome.

This research presents an alternate revenue generation mechanism that could be sustainable for financing HSR construction, operation, and maintenance. The methodology involves identifying key HSR stations, which, after development and improvement, could significantly add value to businesses and real estate growth. Any form of real estate taxes levied on properties surrounding such stations could substantially support the HSR project's funding needs. However, research shows that not every HSR station promotes increased economic activity, and impacts vary based on the station's city. Therefore, it will be useful to know which stations could focus on development and improvement to cause increased economic activity in their surrounding areas for both revenue generation and growth in economic potential.

In this research, a bi-objective optimization problem is posed in conjunction with a Pareto-optimal front framework to identify those key stations. The first objective involves maximizing the revenue potential and the second objective involves maximizing the percentage change in potential accessibility. The Pareto-optimal front is defined as a set of non-dominated solutions that is considered as optimal. A non-dominated solution in the Pareto-optimal front is a set of objective function values that are not dominated by values of any other pair of objective functions in the solution set.

With 28 proposed California HSR stations used as a sample set, it was observed that the four proposed HSR stations in Fullerton, Millbrae-SFO, San Francisco Transbay Terminal, and San Diego would be excellent candidates for development. Their development could increase the economic vitality of surrounding businesses. These four stations are located at some of California's densest employment hubs and would become well connected once HSR becomes fully operational in the state.

Out of the four non-dominated stations proposed as solutions, the proposed Millbrae-SFO and San Francisco Transbay Terminal stations are located very close to each other. The San Francisco Transbay Terminal station is proposed to be the terminal station of the HSR line. The proposed

Millbrae station already serves Caltrain and Bay Area Rapid Transit (BART) modes near San Francisco International Airport (SFO). The enhanced transit connectivity provided by the HSR station at Millbrae-SFO will encourage surrounding station area development and promote economic growth in conjunction with other existing transportation modes. Currently, a draft environmental document is underway for determining the Millbrae-SFO station location and alignment. The proposed HSR station in San Francisco is the Salesforce Transit Center (STC), which will serve 11 transportation systems. This proposed HSR station in San Francisco has been environmentally cleared by the Transbay Joint Powers Authority (TJPA). Fullerton's proposed station is currently under evaluation on the Los Angeles to Anaheim HSR project section. The proposed HSR station of San Diego is to be located close to the San Diego International Terminus. Thus, the station will facilitate access to the air transportation mode once HSR becomes operational.

The findings could serve as valuable information prompting California HSR authorities to develop the four aforementioned vital stations that could generate an alternate funding source for an HSR project facing funding challenges.

I. Introduction and Background

High-speed rail (HSR) promotes a region's economic integration by enhancing competitiveness and achieving greater cohesion among people (Cheng et al., 2015). However, in the United States, the full potential of HSR still needs to be realized. Current progress in HSR development in the country has either been slow or uncertain. Examples of the HSR development have been in Texas that would connect Houston and Dallas and the California HSR program proposed to connect at least 26 different urban and rural cities (Geffen, 2018). Findings suggest that the HSR is needed in the United States to connect economically disadvantaged regions (Chandra and Vadali, 2014). However, major challenges need to be overcome in support of HSR projects, including planning its alignment without disputes/lawsuits, land acquisitions, environmental clearance, etc. Among these challenges, the availability of capital funds for HSR infrastructure construction, operations, and maintenance is often the most crucial.

Funding for large-scale infrastructure projects often involves partnerships with public entities at the local, regional, and/or national levels. However, unforeseen challenges thwart full-fledged funding of projects that use public funds. This is because public funds are often politically invested, temporally limited, tax-dependent, and sensitive to the public agency's economic health. As a result, funds provided primarily by public agencies often run short and hinder the timely completion of a large-scale infrastructure project. California's HSR project is one such example (Government Technology, 2020). Therefore, alternative sources of funding for HSR projects are always welcome.

The HSR serves as a fast transportation mode facilitating mobility, promoting safety, and increasing intermodal connectivity—very conducive for thriving businesses around its stations in cities (Diao, 2018). Several studies show that upgrades and improvements of rail-related infrastructure (such as station development, grade separations, etc.) have increased real estate value, and businesses prosper around those improvements (Cao and Porter-Nelson, 2016; Aldrete et al., 2018). However, guidance on determining the optimal location of such improvements to tap into the economic benefits is still missing in the research.

Projects that have limited funding but are deemed important use innovative methods like the Transportation Reinvestment Zone (TRZ), similar to value capture, to pay off capital loans. TRZ has been emerging as an alternative revenue generation tool for managing large-scale transportation improvement projects in states like Texas and elsewhere in the United States (Aldrete et al., 2018).

Several existing studies have evaluated economic growth potential to determine the economic benefits available to a region through rail station and transit-oriented developments (Zhang and Yen, 2020; Murakami and Cervero, 2010; Belzer et al., 2011). A vast majority of studies mention access improvements (as an indicator of economic growth potential) to justify the construction or revitalization of transportation facilities in a region (Gutiérrez, 2001; Chandra and Vadali, 2014).

Several studies provide evidence of the connection between HSR introduction in a region and property values. Hensher and Mulley (2012) examine the impact of accessibility delivered by HSR on land and property values and suggest that HSR can affect land values to varying degrees, including price premiums and price reductions. Earlier, Cervero and Duncan (2002) also identified the varying degrees of benefits and premiums to residential properties, especially for large apartments within a quarter-mile distance of rail transit stations. Another study by Shi and Guo (2009) on the Shanghai South Railway Station in China also concluded that impacts on housing prices were mainly experienced within a mile of proximity to the rail stations.

Literature on heavy rail transit from South Korea, the Netherlands, and Germany has corroborated the positive impacts of rail on residential housing prices (Bae et al., 2003; Debrezion et al., 2010; Brandt and Maennig, 2012). Earlier, Diaz (1999) identified a variety of factors that increase property values around rail transit. One of the factors was the increase in accessibility brought about by the new transit investment. A study by Ryan (1999) showed a relationship between transportation facilities and property values indicated that the use of travel time as a measure of accessibility gave the expected inverse relationships between access to transportation facilities and property values. However, the results for travel distance indicated mixed property value effects. Kilpatrick et al. (2007) examined two situations of transit corridors: one with both access benefits and negatives, and another without access benefit. The study criticized prior studies that only consider the overall impact on affected properties, combining positive public good and the negative externality, of the transit. The findings revealed that mere proximity to the corridor without direct access has a negative effect on housing values.

A study by Chen and Haynes (2015) showed that the Beijing-Shanghai HSR line increased the housing values in medium and small cities; however, negligible impacts were noted in larger capital cities. Edwards (2012) had found similar results for regional Australian towns traversed by the HSR, causing agglomeration benefits and benefits to cities that had a HSR station. Studies on HSR in Taiwan (Andersson et al., 2010) and China's large cities like Shanghai and Minhang (Rungskunroch et al., 2020), along with a Delphi survey by Loukaitou-Sideris et al. (2012), concluded that the impacts of HSR depend on the context and circumstances such as the extent of regional accessibility, city development plans, and so on.

For HSR projects, careful consideration is required during the planning phase because of the financial hurdles and the time it takes to start service and operations. The ability to create a financial plan is essential to the construction of the HSR. Several examples of partnerships involve shared funding commitments between public and private entities for HSR projects. The appeal of socio-economic returns allows governments to provide public funding, while the appeal of financial return allows private investors to provide private funding. The trans-European HSR project had multiple forms of funding measures (such as loans) for their member states from the Trans-European Transport Networks (TEN) budget line, as well as a European Investment Bank (EIB) and European Investment Fund (EIF) loan (Roll and Verbeke, 1998). It is common for the Chinese government to provide financial assistance for HSR infrastructure construction at about

¥600 billion RMB (about \$91 billion USD) per year (Long and Zhang, 2019). As for India, the HSR line that will connect Mumbai and Ahmedabad will cost around ₹977 billion INR (about \$13 billion USD), although the external sponsor, the Japan International Cooperation Agency (JICA), will contribute almost 80% of the project costs (Raghuram and Udayakumar, 2016). The funding for the Taiwan High Speed Rail (THSR) came from two sources: shareholder equity accounted for 20%, and local banking groups took care of the remaining 80% (Cheng, 2010). Table 1 presents a compilation of funding sources for some examples of HSR projects in various European jurisdictions, plus California.

Table 1. Examples of Funding Sources for High-Speed Rail (HSR)

Name	Location	Funding	Amount (USD equivalent)	Year Reported	Reference(s)
Lisbon-Madrid high-speed rail line	Portugal	European Regional Development Fund	€205.1m (\$239.41m)	2016	TPF, 2020
High-speed rail in Italy (Turin-Lyon line)	Italy and France	European Union	40% of €26 billion (\$12.14b)	Mid 2019	Railway Technology, 2020
Brescia High Speed Line	Italy	European Commissioner for Transport	€1.7 billion (\$1.98b)	December 2008	Railway Gazette International, 2020
RailJet	Austria	Liechtenstein (33.75%) and Austria (Österreichische Bundesfinanzierungsagentur, OeBFA)	€63 million [Liechtenstein (\$73.54m)] and the remainder from Austria	2016	International Railway Journal, 2020
High Speed Train Sweden (SJ AB)	Sweden	NIB (Nordic Investment Bank) and the Swedish passenger train operator	€144 million (\$170m)	2020	Nordic Investment Bank, 2020
High-Speed Train Madrid-Castilla La Mancha-Comunidad Valenciana-Region of Murcia	Spain	EU's European Regional Development Fund	\$51.5 million	2007–2013	European Commission, 2020a
HST railway station in Guillemins	Belgium	European Regional Development Fund (ERDF)	\$53.36 million	2009	European Commission, 2020b
California High-Speed Rail (proposed)	CA	American Recovery and Reinvestment Act of 2009 (ARRA)	\$2.5 billion	2009	California High-Speed Rail Authority, 2020c
		Transportation, Housing and Urban Development grant 2010	\$939 million	2010	
		Cap-and-Trade Program	25 percent of the annual proceeds	2014–2030	

II. Research Contribution

Public authorities seek sustained funds to complete large-scale infrastructure projects and studies show that for rail transit, revenue collected from property tax collections from surrounding parcels of rail stations could fulfill the funding needs. However, there is no clear guidance on the procedures that the public authorities can adopt to identify one or more key HSR stations for development and subsequent revenue generation. Guidance is needed to help reduce the loans or debts incurred in the construction of HSR infrastructure. Several competing factors and objectives need to be considered in this process. A multi-objective optimization approach is needed for this purpose.

This research proposes a bi-objective optimization methodology by incorporating the ‘value capture’ and ‘attractiveness’ potential of stations considered for development. A station that would yield the largest values for both the value capture and attractiveness objectives should be selected for development. The value capture potential is estimated by maximizing the revenue potential whereas the attractiveness is calculated by maximizing the percentage change in the station’s potential accessibility. Subsequently, a Pareto-optimal front is constructed by comparing the two objective functions across candidate stations being analyzed. The Pareto-optimal front is the set of non-dominated solutions that is considered as optimal. A non-dominated solution in the Pareto-optimal front is a set of objective function values that are not dominated by values of any other pair of objective functions in the solution set.

Stations that rank highest in revenue collection through value capture and potential accessibility in terms of attractiveness form the front. The methodology can be used to study the case of California’s HSR, which urgently needs to attract businesses around its key proposed stations and increase the revenue potential, and those stations becoming promising as hubs of economic activity. This can motivate continued investment in HSR to promote its completion and its successful operation to benefit Californians.

Substantial resources have already gone into funding the HSR program—almost ten years of effort—and an unfinished project could lead to the loss of jobs that have been sustained by the HSR program. Thus, innovative funding mechanisms that have often proved to be effective for large-scale infrastructure projects could be evaluated for California’s HSR completion. There are limited literatures on tools to leverage funds from evaluating all aspects of economic benefits accrued from HSR construction and completion. Alternative financing sources, such as value capture, can be explored to fund California’s HSR program, which should be achieved through sustainable and innovative means. The findings from this proposed research would help decision makers determine benefits that have not yet been thought of being harnessed if completion of HSR segments is carried out at strategic locations to connect critical stations and making HSR operational. The methodology presented in this research can be used to achieve this purpose. The goal is to determine those HSR stations that could trigger increases in revenue potential and accessibility to generate further economic activity surrounding the stations. The methodology

proposed in this research can also be applied to study other HSR projects that are facing similar funding challenges.

III. Methodology

A proposed HSR station can be analyzed for its potential to produce economic and financial benefits upon development and improvement. These improvements could attract businesses to the vicinity, ultimately leading to land value increase surrounding the station. The expected outcome due to the improvement, therefore, supports the analysis for their revenue generation potential and the extent of economic activity (e.g., employment growth) in their vicinity.

Research shows that not every HSR station promotes increased economic activity, and impacts vary based on the station's city (Chen and Haynes, 2015). Therefore, it will be useful to know which stations could be targeted for development and improvement to cause increased economic activity in the stations' surroundings for revenue generation as well as growth in economic potential. Such an investigation would also provide information on the extent and magnitude of these economic impacts once the key stations become operational for service.

Therefore, in this research, a methodology is proposed to solve a bi-objective optimization problem. The optimization problem seeks to identify one or more HSR stations that have the largest increase in revenue generated from tax collected from the real estate properties surrounding the station and the largest increase in the station's potential accessibility value. However, this might not be possible: i.e., a station could have the largest potential accessibility increase but its revenue generation could lag. For a better understanding of the interaction between the potential accessibility and revenue generation, formulations are developed for each of the two objectives as follows.

3.1 Maximizing Revenue Potential

Value capture is commonly used in Europe and the United States for revenue generation to fund transit (Salon et al., 2019). For example, existing laws in California provide guidance (and regulation) in funding infrastructure projects; a tax increment tool called the Enhanced Infrastructure Financing District (EIFD), similar to value capture, is used for economic development (League of California Cities, 2019). EIFD authorizes a city's legislative body or a county within the state to use tax increment financing (TIF) to finance various infrastructure improvement projects within its jurisdiction. EIFD enables cities to provide a stable funding source for infrastructure projects, exert leverage, and induce private investments. For an HSR station, TIF-based EIFD would depend on tax collected from surrounding properties. Therefore, stations with high property taxes collected from parcels surrounding the station will be favored for development and improvement.

The first objective function for the optimization problem aims to maximize the total property tax from every parcel within a given threshold distance from a station. Thus, the objective function for maximizing revenue generated, $\max(RP_i)$, can be written as

Objective function 1: Maximization of revenue generated

$$\max(RP_i) = \sum_k a_{i,k}^D, \text{ and } k \in P \quad (1)$$

where

$a_{i,k}^D$ = property tax collected from parcel k within a (threshold) distance D from station i

P = total number of parcels.

3.2 Maximizing Accessibility

Accessibility is often used as an indicator for the economic potential of a location (Chandra and Vadali, 2014). In this research, a proposed HSR station's economic potential is assessed using its accessibility, which would entail future growth in surrounding property values. The formula for accessibility adopted in this research is based on the potential accessibility derived by Chandra and Mazin (2020). Therefore, the potential accessibility (PA_i) of a HSR station i is expressed as

$$PA_i = \sum_j \frac{b_{i,j}}{F(I_{i,j})}, \text{ } i \neq j \text{ and } j \in N \quad (2)$$

where

b_{ij} = total riders boarding the origin station i to reach destination station j

N = total number of stations in the rail transportation network

$F(I_{i,j})$ = the impedance function with I_{ij} could be assumed to be the distance, travel time, energy cost etc. and is incurred from a given origin station i to a destination station j .

The function assumes a gravity-based form proposed by Hansen (1959) with a decay parameter α , and thus, the function can be simply written as

$$F(I_{i,j}) = I_{i,j}^\alpha \quad (3)$$

The decay parameter α in Eq. (3) indicates the variation in accessibility between an origin i and a destination j . In other words, the value of the decay parameter determines how far rail commuters would travel from the origin i to destination j using a mode. In practice, the decay parameter needs travel demand data for calibration and estimation. However, as a rule of thumb, the decay parameter value is often assumed to be 1 in the absence of travel demand data (Gutiérrez, 2001).

The accessibility formula shown in Eq. (2) describes a direct relationship between commuters boarding an origin station and traveling to another destination station. In the absence of exact ridership comparison between stations, the number of commuters using the HSR can be assumed to be proportional to the magnitude of employment from industries within a given threshold

distance around its stations. Numerous studies have also shown that transit ridership is proportional to the population size (Liu et al., 2016; Hiramatsu, 2018). Therefore, in the absence of actual ridership data for Eq. (1), population or employment around a station can be used for an approximation.

With the further assumption that the average speed of the HSR is constant between any two stations, the impedance function in Eq. (2) would be directly proportional to the average travel time taken between the stations. Thus, the accessibility for a station in Eq. (2) can be rewritten as

$$PA_i = \sum_j \frac{b_{i,j}}{t_{i,j}^\alpha}, \quad i \neq j \text{ and } i, j \in N \quad (4)$$

where $t_{i,j}$ is the travel time with HSR between the station i (as origin) and station j (as destination) along the rail network.

A station that has the largest value for potential accessibility change (as a percentage) should be selected for development and improvement. This would entail sustained economic growth around the station. In this research, the percentage change in PA_i for a station i before and after an improvement is used to evaluate the largest value of potential accessibility change. The percentage change in PA_i for a station i is calculated as future year PA_i ('after' scenario with the HSR line) minus base year PA_i ('before' scenario without the HSR line), and the difference is divided by base year PA_i ('before' scenario without the HSR line). The percentage change in PA_i is obtained by multiplying the ratio by 100. Therefore, the second objective function maximizing the percentage change in potential accessibility of a station i , $\max(\Delta PA_i)$, is expressed as

Objective function 2: Maximization of potential accessibility

$$\max(\Delta PA_i) = \left(\frac{\sum_j \frac{b_j^{after}}{(t_{i,j}^{after,HSR})^\alpha} - \sum_j \frac{b_j^{before}}{(t_{i,j}^{before,highway})^\alpha}}{\sum_j \frac{b_j^{before}}{(t_{i,j}^{before,highway})^\alpha}} \right) \times 100, \quad i \neq j \text{ and } i, j \in N \quad (5)$$

where

b_j^{after} = employment surrounding a HSR station j after HSR is operational

b_j^{before} = employment surrounding a HSR station j before HSR is operational

$t_{i,j}^{after,HSR}$ = travel time from station i to station j with HSR operational

$t_{i,j}^{before,highway}$ = travel time from a city with station i to a city with station j using highway.

This method of evaluating percentage change in PA_i with normalization is justified in studies by Chandra and Mazin (2020) and Chandra and Vadali (2014). In this way, each station i is evaluated for its attractiveness resulting from the increased employment weighted access to stations.

3.3 Bi-Objective Design

With the two objective functions defined in Eqs. (1) and (5), the next step is to determine stations that satisfy both functions. Therefore, solutions are sought for the bi-objective optimization problem with a general formulation adapted from Brisset and Gillon (2015). Solutions are presented below for the two objective functions:

$$\begin{aligned}
 & \text{maximize } \mathbf{F}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x})] \\
 & \text{subject to: } g_r(\mathbf{x}) \leq 0 \quad r = 1, \dots, n_r \\
 & \quad \quad \quad h_s(\mathbf{x}) = 0 \quad s = 1, \dots, n_s \\
 & \text{with } \mathbf{x} = [x_1, \dots, x_c, \dots, x_n] \text{ and } x_c^l \leq x_c \leq x_c^u
 \end{aligned} \tag{6}$$

where

$\mathbf{F}$ = set of two objective functions

$g_r(\mathbf{x})$ = total n_r inequality constraints involving the variables $\mathbf{x}$ used in revenue potential and percentage change in accessibility

$h_s(\mathbf{x})$ = total n_s equality constraints involving the variables $\mathbf{x}$ used in revenue potential and percentage change in accessibility

$\mathbf{x}$ = vector containing n variables (that make up the revenue potential and percentage change in accessibility formula)

x_c^l = lower limit to the variable x_c

x_c^u = upper limit to the variable x_c .

The variable n is the number of stations as the elements of vector $\mathbf{x}$. A station can be included in the Pareto front if it has particular values for revenue potential and accessibility and dominates other stations for these two values. The following condition should be satisfied:

$$f_i(\mathbf{x}_1) \leq f_i(\mathbf{x}_2) \quad \forall i \in \{1, \dots, m\} \text{ and } \exists j \in \{1, \dots, m\} \mid f_j(\mathbf{x}_1) < f_j(\mathbf{x}_2) \tag{7}$$

A station is considered non-dominated or Pareto-optimal if no other station dominates it across the two objective functions. All non-dominated stations are on the Pareto front. Thus, one or more stations that fall into the Pareto front, as per the condition imposed by Eq. (7), can determine the outcome of the bi-objective optimization approach expressed using Eq. (6). This means a station that falls on the Pareto front qualifies to be selected for improvement and development to satisfy the two objective functions: i.e., maximization of the potential accessibility and revenue generation.

IV. Application of the Method

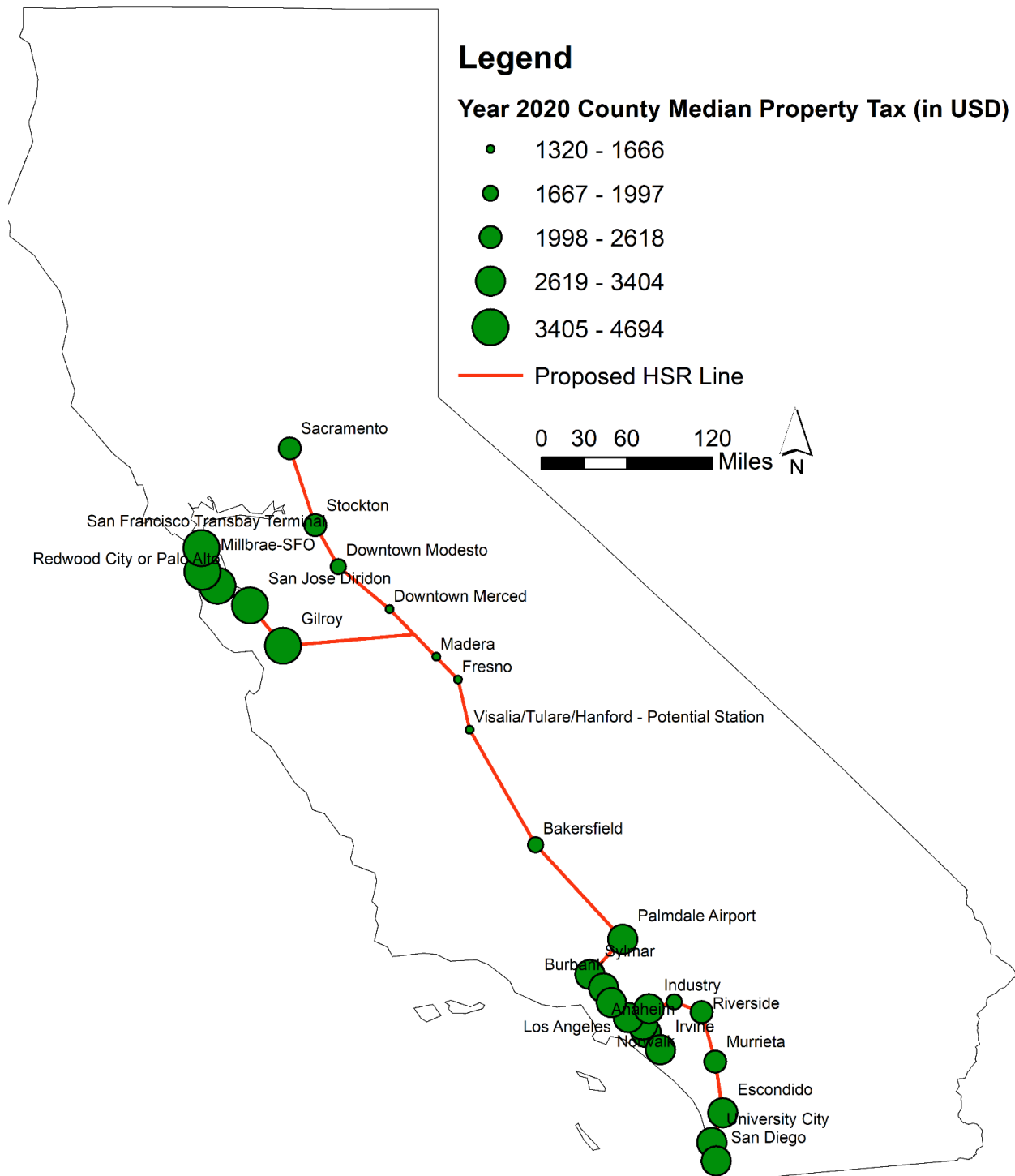
California HSR has recently faced uncertainty in funding from its federal partner, the Federal Railroad Administration (FRA, 2019). In May 2019, the FRA cancelled its entire funding initially intended for the completion of the HSR program. The primary reason cited was the slow progress of the project, and it was also understood as a response to the Governor's announcement that only a short segment section of the HSR line would be completed linking Merced and Bakersfield in the state's Central Valley.

The county property tax is used to evaluate revenue potential as expressed in objective function 1 (i.e. maximizing revenue potential) of Eq. (1). In the absence of actual data needed for parcels' property taxes surrounding the stations, the median property taxes at the spatial level of counties are used as an approximation (data from Tax-rates.org, 2020).

Research shows that economic impacts due to transit improvement are primarily concentrated within a quarter-mile radius from the improvement (Cervero and Duncan, 2002). Therefore, parcels within a quarter-mile radius surrounding a station are considered for the total property tax calculation for objective function 1. The number of parcels surrounding an HSR station is collected from the 2014 California Parcels Dataset from Los Angeles County GIS webpage (LA County GIS, 2020).

The map in Figure 1 shows the proposed California HSR line and the stations with county median property taxes for properties surrounding each station.

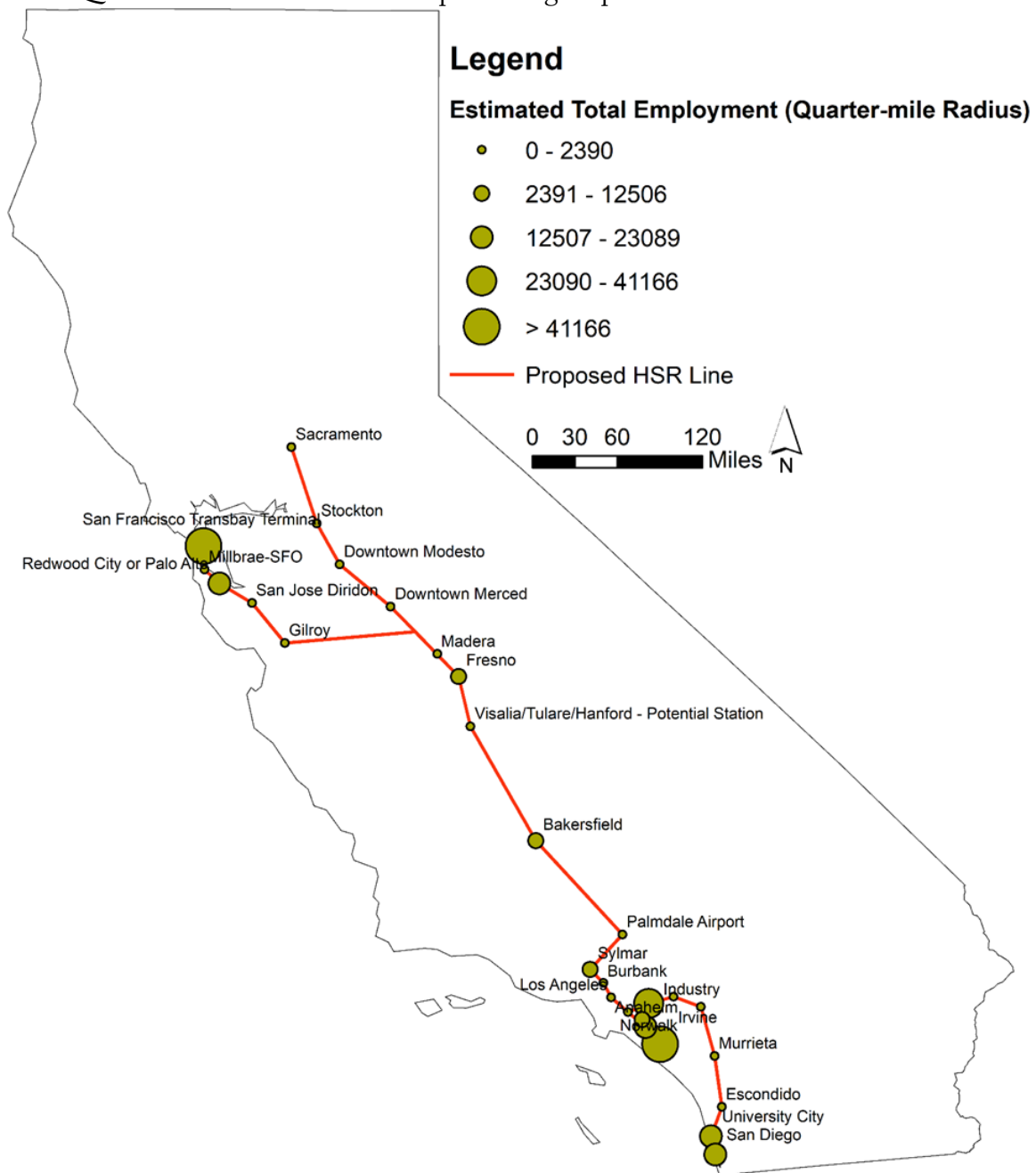
Figure 1. Spatial Distribution of Property Tax Surrounding the Proposed High-Speed Rail Stations in California



Employment data needed for the accessibility calculation in Eq. (5) are obtained from the Longitudinal Employer-Household Dynamics (LEHD) that provides the data years 2002 through 2017 (LEHD, 2020). The employment is extrapolated to 2020 with the anticipated HSR completion by 2022.

The map in Figure 2 shows the spatial variation in total employment for the year 2020 from all the industry sectors within a quarter-mile distance surrounding the proposed stations.

Figure 2. Spatial Distribution of the Estimated Total Employment for the Year 2020 within a Quarter-Mile Radius of the Proposed High-Speed Rail Stations in California



For impedance used in the accessibility formulation, the travel time skim matrix was obtained by dividing the distance between two stations by the assumed average speed of 200 miles per hour for the HSR (California High-Speed Rail, 2020b). The spatial maps for the HSR line and stations

(total 28) shown in Figs. 1 and 2 were created using data from the California High-Speed Rail Authority (California High-Speed Rail, 2020a).

The data collected were finally used in determining the objective functions under Eqs. (1) and (5) and the outcome discussed in the next section.

V. Results and Discussion

The constraints used in the bi-objective optimization are such that both the revenue potential and the percentage change in potential accessibility are assumed to be positive. The analysis of the revenue potential using Eq. (1) shows that the proposed HSR stations of San Diego and San Francisco Transbay Terminal have the highest revenue potential among other proposed HSR stations. This is because the number of parcels within a quarter-mile distance is higher than the corresponding values for the other stations. Thus, if the focus is solely on generating revenue from property taxes under EIFD tax increment strategy, these two HSR stations should be developed and improved. The HSR stations of Millbrae-SFO and San Jose Diridon are the next two stations that should be considered for development and improvement for revenue generation purposes. (The spatial location of these four HSR stations is shown in Fig. 3.)

If the focus is on maximizing the potential accessibility (Eq. (5)), the stations of Fullerton, Millbrae-SFO, Norwalk, and Anaheim should be preferred for development and improvement. These four stations have the largest percentage change in potential accessibility. The high percentage change in accessibility value for these four stations can be attributed to their proximity to other stations with high employment within the quarter-mile distance threshold. (Fig. 4 shows the spatial variation in the percentage change in potential accessibility across the HSR stations.)

After obtaining the outputs from the bi-objective optimization, the authors constructed the Pareto-optimal front. The Pareto-optimal front presented in Figure 5 shows that the four proposed HSR stations of Fullerton, Millbrae-SFO, San Francisco Transbay Terminal, and San Diego stand out as the non-dominated solutions. These four stations are located at some of California's densest employment hubs and would become well-connected once HSR becomes fully operational in the state. Out of the four non-dominated stations, the proposed Millbrae-SFO and San Francisco Transbay Terminal stations are located very close to each other. The San Francisco Transbay Terminal station is proposed to be the end station of the HSR line.

The proposed Millbrae station already serves the Caltrain and Bay Area Rapid Transit (BART) modes near San Francisco International Airport (California High-Speed Rail Authority, 2020d). With the enhanced transit connectivity provided by the HSR station at Millbrae-SFO, the station will encourage surrounding station area development and promote economic growth in conjunction with other existing transportation modes. Currently, a draft environmental document is underway for determining the Millbrae-SFO station location and alignment.

The proposed HSR station in San Francisco is the Salesforce Transit Center (STC), which will serve eleven transportation systems (California High-Speed Rail Authority, 2020e). The current status of this proposed HSR station in San Francisco is that it has been environmentally cleared by the Transbay Joint Powers Authority (TJPA).

Fullerton's proposed station is currently under evaluation on the Los Angeles to Anaheim Project Section (California High-Speed Rail Authority, 2020f). The proposed San Diego HSR station is to be located close to the San Diego International Terminus (California High-Speed Rail Authority, 2020g). Thus, the station will facilitate access to the air transportation mode once HSR becomes operational.

Figure 3. Spatial Variation in the Revenue Potential of Proposed HSR Stations in CA

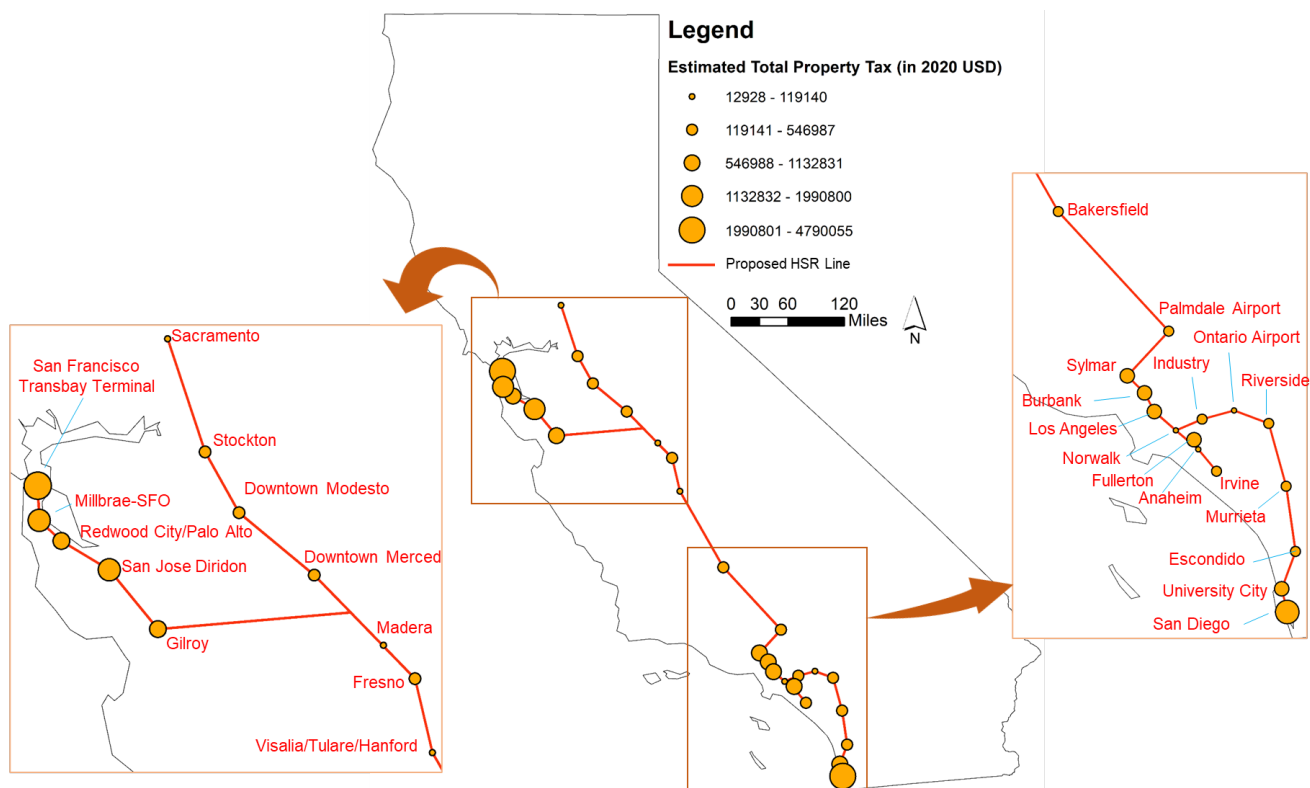


Figure 4. Spatial Variation in the Percentage Change in Potential Accessibility of the Proposed HSR Stations in CA

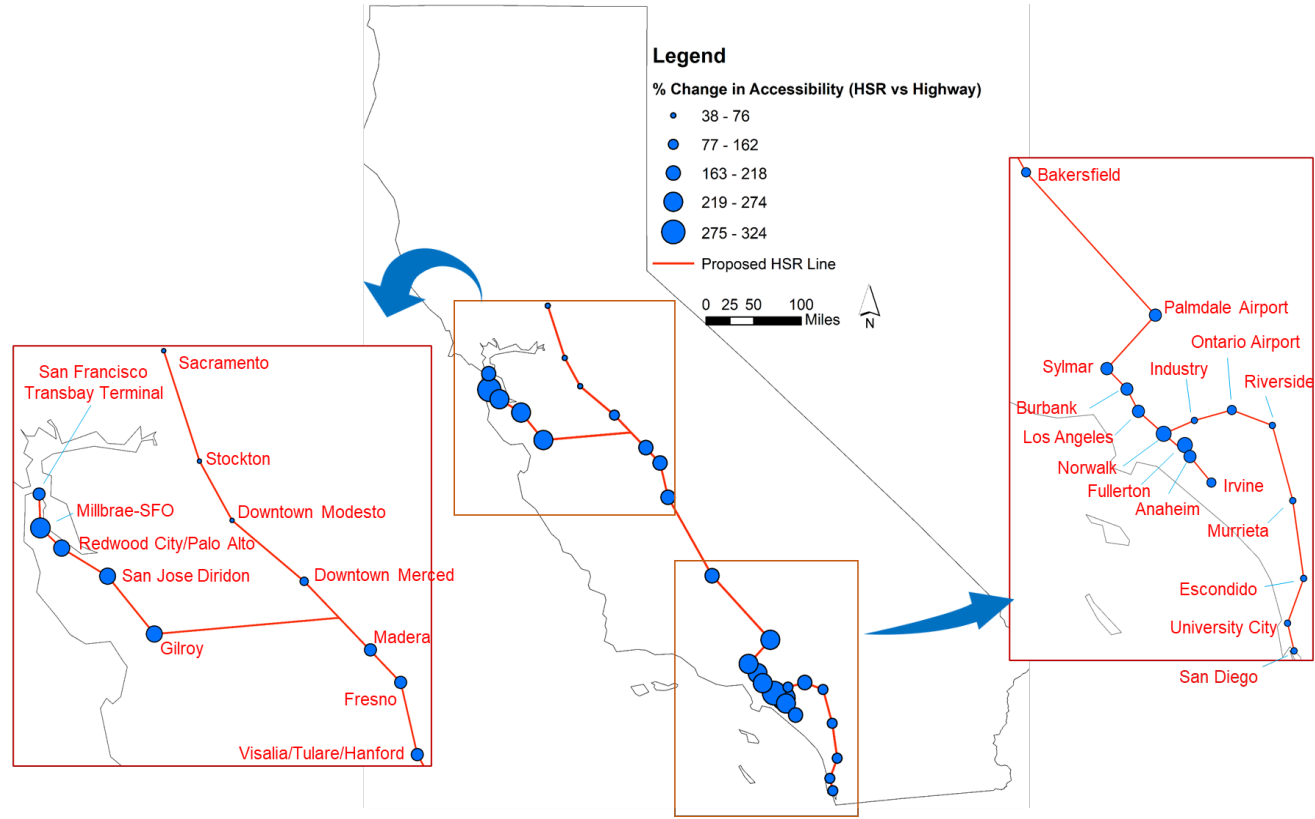
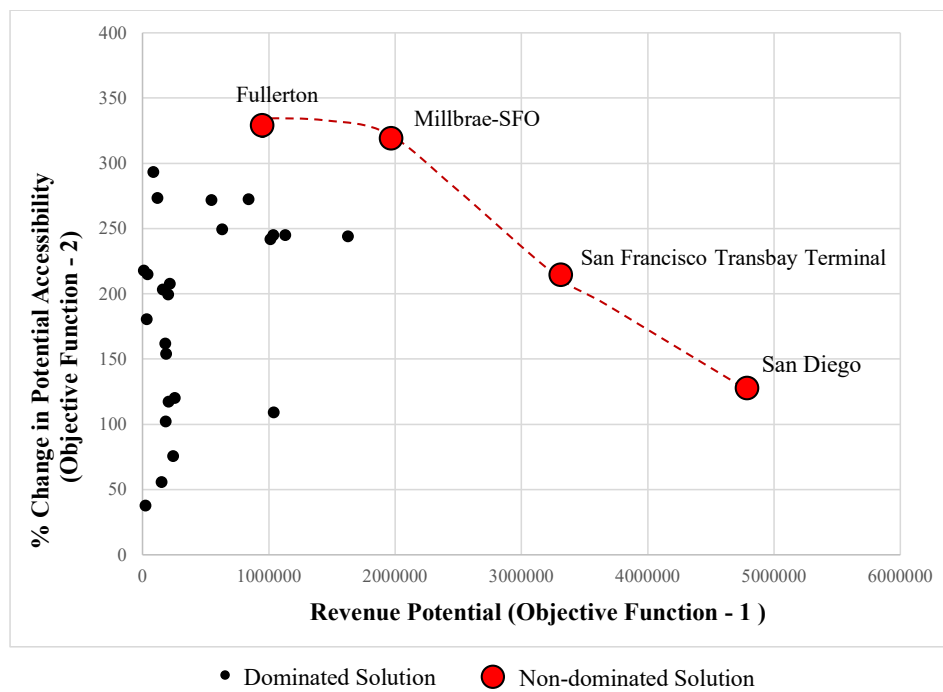


Figure 5. Pareto-Optimal Solution Representation for a Bi-Objective Optimization Problem



VI. Summary and Conclusions

The HSR serves as a fast surface transportation mode facilitating mobility between two spatially separated regions with stations. HSR stations serve as points of connectivity and are selected to be hotspots of economic activities. With the purpose being to spur economic growth with HSR, authorities upgrade and improve existing station infrastructure. Economic growth includes an increase in real estate value and businesses surrounding the stations. However, developing, upgrading, or improving an HSR station requires funds that are often limited. Therefore, various funding options should be explored to promote the development and upgrade of HSR stations to standards that can attract businesses in the surrounding area.

In this research, a methodology is presented that can be deployed to identify stations that could have the potential to attract economic activities after upgrades. It involves using a bi-objective optimization framework for identifying one or more HSR stations from a list of candidates for development. The objective functions maximize the revenue generated from tax collected from the station's surrounding real estate properties and the station's potential accessibility. Subsequently, the solution is presented by constructing a Pareto-optimal front with key proposed stations that decision-makers can focus on for development and upgrade.

Using California's 28 HSR stations as an example, the Pareto-optimal front showed that the four proposed HSR stations of Fullerton, Millbrae-SFO, San Francisco Transbay Terminal, and San Diego would be the best suited for development to increase economic vitality in their surroundings. These four stations stand out as the non-dominated solution of the constructed Pareto-optimal front. The finding could serve as valuable information for California HSR authorities to develop these four key stations to generate an alternate funding source for the HSR project facing funding challenges.

Thus, this research provides useful guidance that HSR authorities and stakeholders can use to identify stations with high economic growth and accessibility potential for a region. Alternatively, the knowledge developed from this research might also help stimulate real estate value increase and business growth surrounding a station by improving its accessibility to other stations on the HSR line.

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Exhibit SR-7

Funding the High-Speed Rail



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RESEARCH ARTICLE

Funding the High-Speed Rail: A Case Study of the California Project

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Abstract:

Introduction:

High-speed rail (HSR) projects are considered high-stake investments— both in construction, operation, and maintenance. Thus, HSR requires financing from various sources – public, private or a mix of both, as the case maybe. Traditional funding for HSR has usually been the public funds through various bonds and taxes. However, public funds are not easy to come by and are usually subjected to strict audits and reviews. Therefore, alternative funding sources are always sought after, especially those that are sustainable.

Methods:

A multiobjective optimization problem is proposed in this paper encompassing revenue generation, accessibility maximization and workforce-centric access measure (WAM) consideration for financing HSR development costs and justifying its purpose. The procedure identifies key HSR stations that could be developed and improved to positively increase revenue collected through taxes on surrounding properties that appreciated due to the HSR, or the appreciation that either occurred through accessibility increase or WAM considerations from the HSR.

Results and Discussion:

Such key stations that score high in one or more of these considerations could be considered for further development by policymakers and planners. Based on the multiobjective optimization problem proposed in this paper, and with the California HSR as an example, the three proposed stations of San Francisco Transbay Terminal, University City and Downtown Merced are recommended to be developed and improved to attract economic growth with the HSR line.

Conclusion:

The findings could help authorities focus on these stations (and similar others) for a sustainable means of at least partially funding California's HSR project.

Keywords: High speed rail, High-stake, Audit, Workforce-centric access measure, Multi objective, Revenue generation.

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1. INTRODUCTION

Despite the benefits being realized from high-speed rail (HSR) in terms of connecting people and influencing economic activity [1], its development in the United States is still currently 'a work-in-progress' and lies at varying stages, and somewhat slow and uncertain. There are examples of the HSR, such as in Texas, that is still in the proposal stages and is aimed to connect Dallas and Houston, while the California HSR program, which had temporary funding issues, has been ongoing and, once completed, will connect various urban and rural cities [2]. Various motivations are behind the idea of pursuing HSR developments in the United States, and the primary purpose is to connect the regions that need mobility as

well as those that need economic [3]. Nevertheless, the road to a complete HSR line in the country has been an uphill one with hurdles that primarily have risen due to reasons including funding crunch, land acquisition issues, environmental clearances, and planning. Out of all of these challenges, funding shortage for HSR construction has been the primary and critical factor that has held smooth progress. The basis of funding HSR projects lies in an effective investment from public agencies in the United States; however, these projects turn out to be large-scale infrastructure improvement endeavors encompassing various counties, possibly states of the country. Thus, various local, regional, and national interests become important when such projects are undertaken. With all of this, it makes sense to involve private players to contribute to offset the initial construction costs [4].

When it comes to using solely public funds, there are

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several barriers that need to be overcome, including those that are political in nature. Politically motivated investments in large-scale projects are quite common but are also primarily derived from public taxes and, more importantly, dependent on the funding agencies' financial health. These have been the reasons for California's HSR, which will connect almost every populated city of the state [5]. Therefore, alternative means of funding HSR projects with the right goal and motivation are always a common pursuit by policymakers, investors, and researchers.

There are key benefits of the HSR, and one of these includes fast connectivity to remote locations, especially for congested cities with vehicular volumes. HSR also promotes safe transportation of people, and if integrated with other modes, it enhances the intermodal transportation network system. These benefits are highly beneficial for businesses that thrive around the HSR stations [6]. In general, it has been shown that infrastructure and operational improvements in rail transportation lead to an increase in property values around the stations and also encourage businesses and industries to set-up their locations close to the improved stations [7, 8]. However, there is limited knowledge in research on how to determine such stations that can be improved to attract businesses and generate investments to further support HSR construction and maintenance.

In states like Texas, alternative financing schemes for funding infrastructure projects have been probed, and these are popularly known as the Transportation Reinvestment Zone (TRZ) – which is an alternative means of generating revenue for public agencies and somewhat resembles the concept of value capture [8]. TRZ has been hailed as an innovative method that can be deployed by local public agencies to pay off capital loans.

Economic growth has been studied for improvements resulting from rail station developments and often from the perspective of transit-oriented development [9 - 11]. Various approaches that have been adopted also consider accessibility as a potential economic indicator to support transportation-related improvements and construction in a region [3, 12].

Similarly, the economic aspect of HSR introduction has also been justified by associating the resulting transport development with the new HSR in a region with surrounding property values in the vicinity of the HSR. For example, it is understood that HSR can impact property and land values in a region variably [13]. In an earlier study, both Cervero and Duncan [14] realized benefits and premiums for large apartments around rail stations. These benefits to apartments were for those that were contained within a threshold distance of a quarter mile to the stations. However, when it came to specifically housing prices, noticeable impacts were noted for those within a mile distance from the rail stations of the Shanghai South Railway Station in China [15].

Global examples from other rail transit studies of the Netherlands, South Korea, and Germany show similar findings of a positive correlation between residential housing prices and rail transit [16 - 18]. A very early study by Diaz [19] showed that investments that result in a new transit increase property

value because of increase in accessibility to locations.

In accessibility studies, Kilpatrick *et al* [20] examined two kinds of scenarios for transit corridors. In one scenario, both the pros and cons of accessibility were accounted for, and the second scenario was studied without considering the access benefits. The findings showed that only being near to a transit corridor, presumably with high accessibility, was not sufficient to bring about positive impacts on the property values. On the contrary, being close to the corridor yet without a direct access can cause housing values to decrease. With a focus on the Beijing-Shanghai HSR line, Chen and Haynes [21] showed a negligible impact on the housing values for the large cities, while there was a significant increase in the values for the medium and small cities. Findings like this were noted in the study by Edwards [22] for the towns in Australia with the proposed HSR. Results showed agglomeration benefits to cities with HSR stations.

A study on the variety of other factors that are considered important in facilitating the HSR's impact include circumstances and context of city development plans, regional accessibility, *etc.* These factors have been noted for the HSR of Taiwan and those in the Chinese cities of Shanghai and Minhang [23, 24]. Keeping these aspects of the HSR in mind, the HSR clearly has to be affordable by virtue of being accessible [25].

2. METHODOLOGY

Multiobjective optimization is a standard procedure to identify optimal solutions to problems. This research presents a multiobjective optimization problem and a solution in identifying one or more HSR stations that can be further developed to potentially offset costs involved in HSR project costs. The optimization problem analyzes the possibility of property value increase, changes in accessibility and workforce-centric access measure that can motivate and attract businesses surrounding the HSR stations. To the best of our knowledge, this is the first-ever study of HSR that evaluates each station of a HSR line by looking at various aspects of HSR justification that involves the workforce across age, earnings, industry distributions, race, ethnicity, educational attainment, and sex.

The basis of developing the multiobjective optimization problem is that while a station might have the largest revenue potential increase from the surrounding property values, the same station could experience the lowest accessibility increase or the highest workforce-centric access measure (WAM) value. To understand how these three objectives interplay, the multiobjective optimization problem can be successfully deployed [26].

2.1. Revenue Potential Formulation

The revenue, RP_i , generated can be written as,

$$\max RP_i = \sum_k a_{i,k}^D, \text{ and } k \in P \quad (1)$$

where,

$a_{i,k}^D$ = property tax collected from a parcel k that is within a

(threshold) distance D from station i and P is the total number of parcels.

2.2. Percentage change in accessibility

The formula for accessibility adopted in this research is based on the potential accessibility derived by Chandra and Mazin [27]. The potential accessibility (PA_i) of an HSR station i is expressed as:

Therefore, the second objective function of maximizing the potential accessibility of station i , $\max(\Delta PA_i)$, is expressed as:

$$\max(\Delta PA_i) = \left\{ \frac{\sum_j \frac{b_j^{\text{after}}}{(t_{i,j}^{\text{after, HSR}})^a} - \sum_j \frac{b_j^{\text{before}}}{(t_{i,j}^{\text{before, highway}})^a}}{\sum_j \frac{b_j^{\text{before}}}{(t_{i,j}^{\text{before, highway}})^a}} \right\} \times 100, \quad i \neq j \text{ and } i, j \in N \quad (2)$$

where,

b_j^{after} = employment surrounding an HSR station j 'after' HSR is operational

b_j^{before} = employment surrounding an HSR station j 'before' HSR is operational

$t_{i,j}^{\text{after, HSR}}$ = travel time from station i to station j with HSR operational

$t_{i,j}^{\text{before, highway}}$ = travel time from a city with station i to a city with station j using a highway

2.3. Workforce-centric Access Measure

The Workforce-centric Access Measure (WAM) provides an estimate of the HSR's ability to address mobility needs across the working class near its stations. Various variables can be WAM-related and could include age, earnings, industry distributions, race, ethnicity, educational attainment, and sex. The WAM_i for a station i can be formalized as,

$$WAM_i = \sum_{j=1}^n H_{i,j}^d \quad (3)$$

where,

$H_{i,j}^d$ = total workforce belonging to a class or category type j working within a threshold distance d around a station i . Typically, this threshold distance can be assumed to be walking distance around the HSR station i . The various categories that can be considered in the WAM could include age, earnings, industry distributions, race, ethnicity, educational attainment, and sex. These variables are considered important for US economic studies – particularly for businesses and industries [28].

2.4. Multi-objective Optimization Design

There are several established methods for determining a non-dominating solution for multi-objective optimization problem. The steps leading to each of the methods differ, thereby leading to differences in the nondominated solution. Wang and Rangai [29] develop a summary of the popular methods, and we select the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) for our purpose [30].

The steps involved in the TOPSIS algorithm are as follows:

Step 1: A normalized objective matrix with m rows (each row standing for a value of an objective function) and n columns (each column stands for an objective function) is constructed. The matrix consists of objective function values.

$$F_{ij} = \frac{f_{ij}}{\sqrt{\sum_{i=1}^m f_{ij}^2}} \quad (4)$$

where,

f_{ij} stands for the i^{th} value of the objective function j ,

F_{ij} is the normalized objective function

m is the number of objective function values, and n is the number of objective functions

Step 2: The normalized function F_{ij} is weighted using w_j and defined using v_{ij} as, $v_{ij} = F_{ij} w_j$

Step 3: Define ideal solutions (positive, A^+ and negative, A^-) as follows:

$$A^+ = \left\{ \left(\max_i (v_{ij}) \mid j \in J \right), \left(\min_i (v_{ij}) \mid j \in J' \right) \mid i \in 1, 2, \dots, m \right\} \\ = \{ v_1^+, v_2^+, v_3^+, \dots, v_j^+, \dots, v_n^+ \}$$

And

$$A^- = \left\{ \left(\min_i (v_{ij}) \mid j \in J \right), \left(\max_i (v_{ij}) \mid j \in J' \right) \mid i \in 1, 2, \dots, m \right\} \\ = \{ v_1^-, v_2^-, v_3^-, \dots, v_j^-, \dots, v_n^- \}$$

With J defined as a set of objectives that need to be maximized, and J' as objectives that need to be minimized for $\{1, 2, 3, \dots, n\}$.

Step 4: Compute the distance (Euclidean) between each solution v_{ij} and the ideal solution (both for the positive and negative) as follows:

$$S_{i+} = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^+)^2}, \quad i = 1, 2, 3, \dots, m$$

$$S_{i-} = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^-)^2}, \quad i = 1, 2, 3, \dots, m$$

Step 5: Calculate the closeness C_i as:

$$C_i = \frac{S_{i-}}{S_{i-} + S_{i+}}$$

The optimal solution is determined by the largest C_i calculated above.

3. APPLICATION EXAMPLE

While some uncertainties prevail in funding the California HSR from the federal agencies [31], the situation can be considered ideal for demonstration of the methodology proposed in this paper – simply because there is a need for the HSR authorities to justify the purpose of the overall project and to procure alternative means of funding. Any lag in project

funding for California's HSR would continue to lead to the increased price of material, labor costs/availability and fast drying of funds. Thus, while innovative means of revenue generation should be found, there is also a societal need to serve as much workforce as possible for the State's economic growth by access to the HSR. The three proposed objectives in this research can help identify key stations that can trigger economic activity surrounding them that may include increase in property values, employment opportunities and connectivity.

The data for the county median property tax has been used to evaluate the revenue potential (data source: Tax-rates.org [32]). In addition, only those properties from parcels are included that are within a quarter-mile distance of a station.

Adopting this threshold distance has been recommended by Cervero and Duncan [14]. The spatial data for the parcels were obtained from the 2014 California Parcels Dataset from Los Angeles County GIS webpage [33]. The map in Fig. (1) shows the spatial location of the stations for the California HSR line, along with the spatial distribution of the county median property taxes for properties surrounding each station. The map in Fig. (2) shows the total employment distribution surrounding the stations for a quarter-mile distance. The data were originally available from the Longitudinal Employer-Household Dynamics [28] and extrapolated for 2020 to the anticipated HSR completion by 2022. Both the maps in Figs. (1 and 2) have been adopted from Chandra *et al* [26].

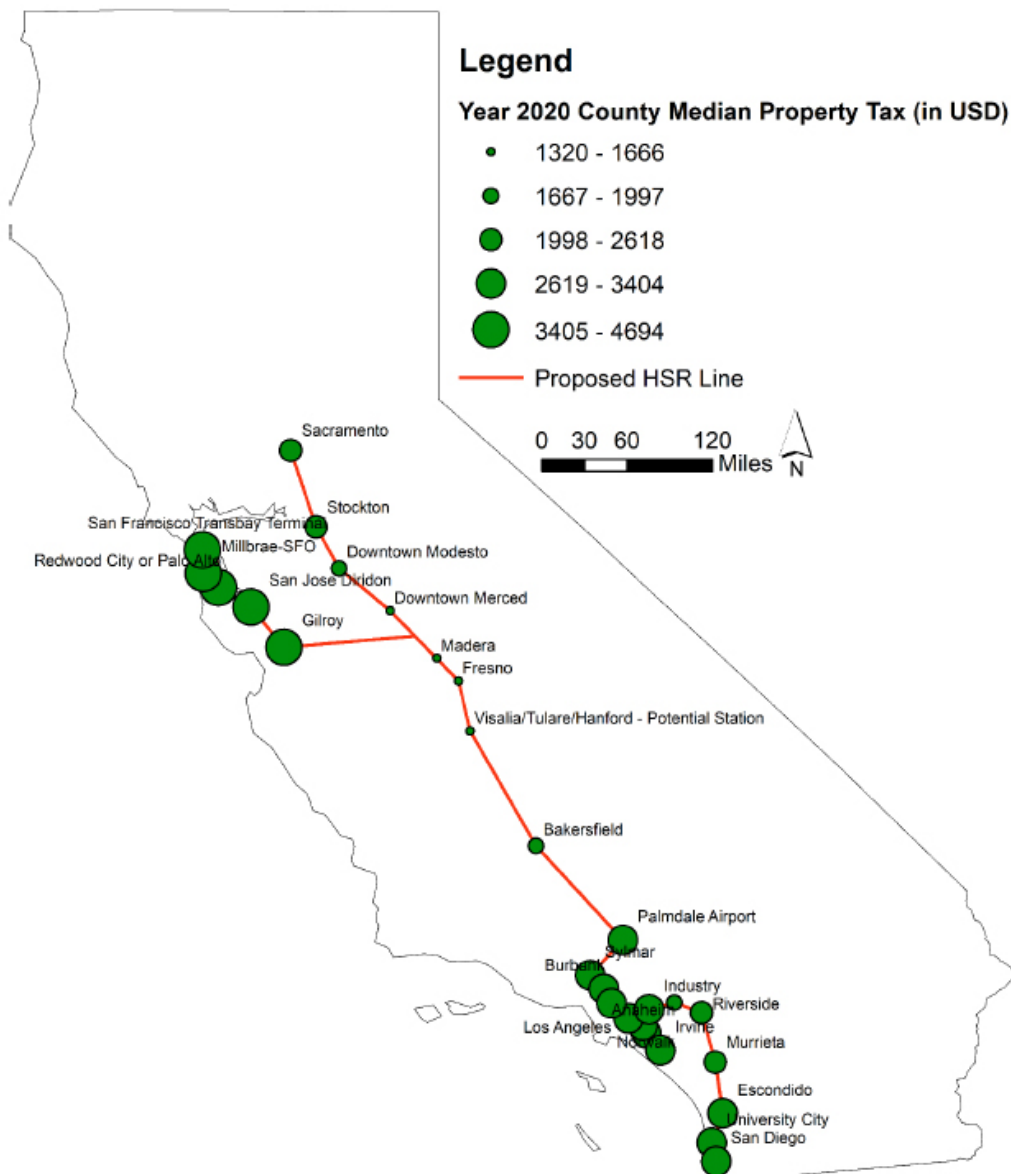


Fig. (1). Spatial distribution of the median county property tax year 2020 for each station.

The data needed for WAM calculations are also obtained from the LEHD and extrapolated for 2020. The variables and their definitions used in the calculation are summarized in Table 1. The variables are widely used for economic measures that underlie the significance of the workforce for California's businesses and industries and are presented as the North American Industry Classification System (NAICS) for the industry type.

For the calculation of accessibility, the impedance used has been derived from the travel time skim matrix that was constructed for the mutual distance between stations and the proposed speed of 200 miles per hour for the HSR [34]. The spatial maps for the HSR line in Figs. (1 and 2) show a total of 28 proposed stations in cities that were obtained from the California High-Speed Rail Authority [35].

Table 1. Workforce access measure classifications and variables of interest.

Workforce Classification by	Variable
Age	Number of jobs of workers age 29 or younger
	Number of jobs for workers age 30 to 54
	Number of jobs for workers age 55 or older
Income	Number of jobs with earnings \$1250/month or less
	Number of jobs with earnings \$1251/month to \$3333/month
	Number of jobs with earnings greater than \$3333/month
Industry-type	Number of jobs in NAICS sector 11 (Agriculture, Forestry, Fishing and Hunting)
	Number of jobs in NAICS sector 21 (Mining, Quarrying, and Oil and Gas Extraction)
	Number of jobs in NAICS sector 22 (Utilities)
	Number of jobs in NAICS sector 23 (Construction)
	Number of jobs in NAICS sector 31-33 (Manufacturing)
	Number of jobs in NAICS sector 42 (Wholesale Trade)
	Number of jobs in NAICS sector 44-45 (Retail Trade)
	Number of jobs in NAICS sector 48-49 (Transportation and Warehousing)
	Number of jobs in NAICS sector 51 (Information)
	Number of jobs in NAICS sector 52 (Finance and Insurance)
	Number of jobs in NAICS sector 53 (Real Estate and Rental and Leasing)
	Number of jobs in NAICS sector 54 (Professional, Scientific, and Technical Services)
	Number of jobs in NAICS sector 55 (Management of Companies and Enterprises)
	Number of jobs in NAICS sector 56 (Administrative and Support and Waste Management and Remediation Services)
	Number of jobs in NAICS sector 61 (Educational Services)
	Number of jobs in NAICS sector 62 (Health Care and Social Assistance)
	Number of jobs in NAICS sector 71 (Arts, Entertainment, and Recreation)
	Number of jobs in NAICS sector 72 (Accommodation and Food Services)
	Number of jobs in NAICS sector 81 (Other Services [except Public Administration])
	Number of jobs in NAICS sector 92 (Public Administration)
Race	Number of jobs of workers, Race: White Alone
	Number of jobs of workers, Race: Black or African American Alone
	Number of jobs of workers, Race: American Indian or Alaska Native Alone
	Number of jobs of workers, Race: Asian Alone
	Number of jobs of workers, Race: Native Hawaiian or Other Pacific Islander Alone
Ethnicity	Number of jobs of workers, Race: Two or More Race Groups
	Number of jobs of workers, Ethnicity: Not Hispanic or Latino
Educational Attainment	Number of jobs of workers, Ethnicity: Hispanic or Latino
	Number of jobs of workers, Educational Attainment: Less than high school
	Number of jobs of workers, Educational Attainment: High school or equivalent, no college
	Number of jobs of workers, Educational Attainment: Some college or Associate degree
Sex	Number of jobs of workers, Educational Attainment: Bachelor's degree or advanced degree
	Number of jobs of workers, Sex: Male
	Number of jobs of workers, Sex: Female

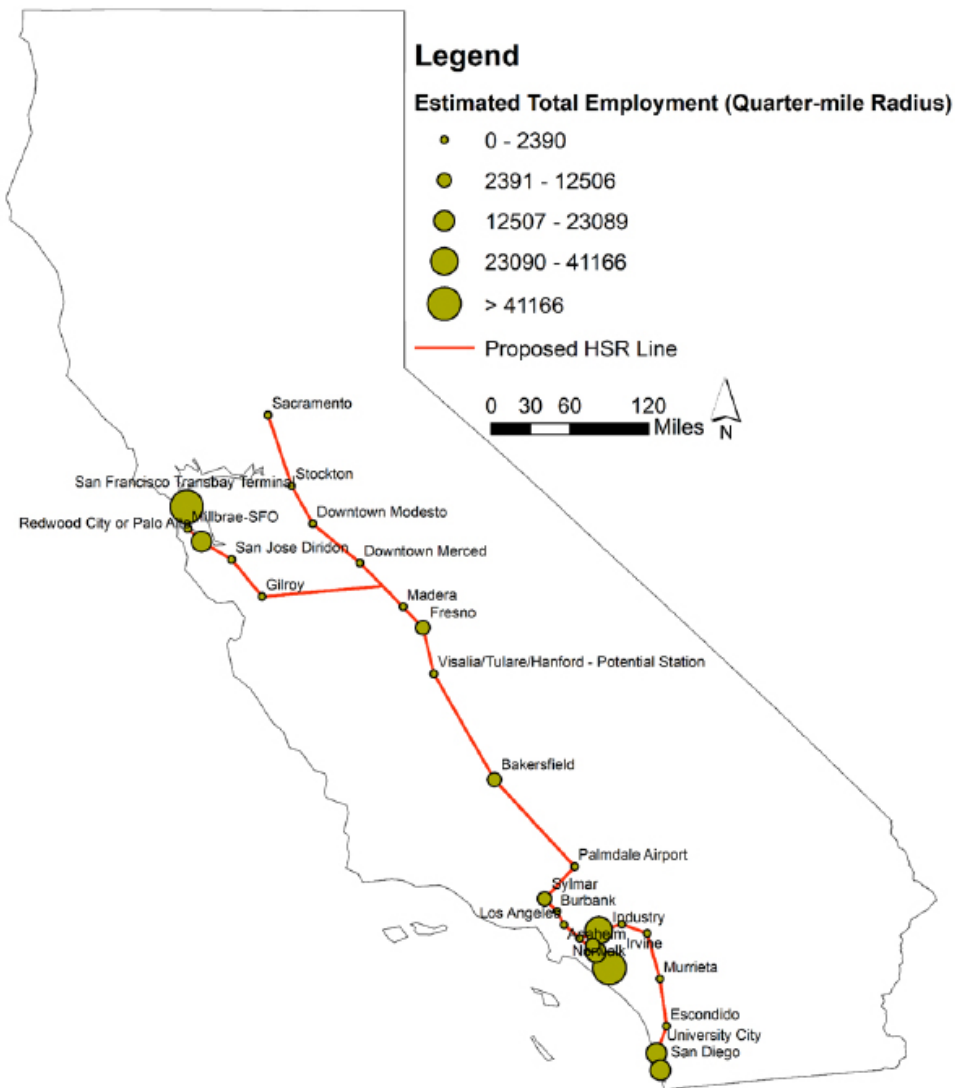


Fig. (2). Spatial distribution of total employment for the year 2020 for each station.

4. RESULTS AND DISCUSSION

The TOPSIS algorithm presented earlier was applied to determine the key California HSR stations for the three objectives. For the analysis that was carried out with the objective of maximizing revenue potential, the San Diego and San Francisco Transbay Terminal stations should be considered important for focus and development by the policymakers and planners. These two stations have the largest potential for generating revenue from the property tax collected from their surrounding parcels. The two HSR stations of Millbrae SFO and San Jose Diridon and Millbrae SFO are the next two stations that could also yield a high revenue if developed and improved.

With the objective of maximizing accessibility, the HSR stations of Millbrae-SFO, Fullerton, Norwalk, and Anaheim score high in percentage change in potential accessibility and thus, are the key HSR stations that need to be improved or

developed. These four stations have a high employment concentration surrounding them.

With the WAM, the stations of Sacramento, San Jose Diridon, San Francisco Transbay Terminal, San Diego, and Redwood City/Palo Alto were found as key stations for development and improvement across age, income, industry, race, ethnicity, educational attainment, and sex classifications, respectively.

Considering all the three objectives together, weighted equally, the three key stations of San Francisco Transbay Terminal, University City and Downtown Merced are recommended for development and improvement to attract economic growth with HSR in California. Therefore, these three stations are found to be optimal considering revenue generation potential, maximizing accessibility, and increasing workforce-centric access measure. The findings have been presented in Table 2.

Table 2. Workforce-centric access measure classification and key stations.

Workforce Classification	Key Station for Improvement
Age	Sacramento
Income	San Jose Diridon
Industry	San Francisco Transbay Terminal
Race	San Diego
Ethnicity	Redwood City/Palo Alto
Educational Attainment	Redwood City/Palo Alto
Sex	Redwood City/Palo Alto

CONCLUSION

The HSR is one of the fast modes for transportation mobility. HSR is known to connect regions that are spatially separated, yet with all of these advantages, construction, operation and maintenance of HSR require sustainable funding sources. Funding is, however, a challenge as public agencies, who are usually the primary stakeholders responsible for HSR projects, seek public support in the form of taxes. There is a need to better understand avenues and leverage them to justify the purpose of HSR construction in providing an important mobility option. This research explores three such objectives in the form of revenue generation, accessibility maximization and workforce-centric access measure to support HSR projects. This paper specifically explores the option of developing and improving key HSR stations that are promising in satisfying the three objectives. The key stations would show indications of high economic activity, which can be leveraged to fund or offset costs incurred in HSR construction, operation and maintenance. With the multiobjective optimization methodology undertaken in this research, an established algorithm called the TOPSIS is applied to identify the key stations. In this current research, multiobjective optimization helps identify critical stations for further development and improvement. This is undoubtedly a concern that investment in underserved communities could be less. However, our goal in this paper was mainly to focus on actions that could generate much-needed revenue for the HSR authorities.

The findings of this research suggest that the with the objective of maximizing just the revenue potential, the San Diego and San Francisco Transbay Terminal stations should be considered important for focus and development. When maximizing accessibility for the second objective, the HSR stations of Millbrae-SFO, Fullerton, Norwalk, and Anaheim score high in percentage change in potential accessibility and are the key HSR stations that need to be improved or developed. With a focus on a workforce-centric access measure (WAM), the stations of Sacramento, San Jose Diridon, San Francisco Transbay Terminal, San Diego, and Redwood City/Palo Alto were found as key stations for development and improvement across age, income, industry, race, ethnicity, educational attainment, and sex classifications.

Considering all the three objectives together, and weighing each equally, the stations of San Francisco Transbay Terminal, University City and Downtown Merced are recommended for development and improvement to attract economic growth with HSR in California.

The findings in this research suggest that developing and improving stations with various objectives could serve as economic incentives for businesses to thrive and grow near select California HSR stations. The increase in economic activities surrounding these stations could mean alternative sources of funds that are sustainable through property taxes and businesses, which can be utilized to fund and support HSR projects.

It must be noted that our current research is not focused on addressing the inequality that the CA HSR line might bring, and this is our research limitation but certainly forms the goal of our future research. We also believe there could be other methods of revenue generation, such as bonds, other than taxation [8]. However, taxation is a primary source of funding infrastructure projects when property values near the HSR station increase and thus, appear justifiable in investment near high-yield property appreciations near specific HSR stations from an institutional point-of-view. Thus, the methodology proposed in this paper can serve as a very valuable reference for financing HSR projects for transport connectivity by public and/or private agencies elsewhere in other regions of the United States.

LIST OF ABBREVIATIONS

WAM	=	Workforcecentric Access Measure
HSR	=	High-Speed Rail

CONSENT FOR PUBLICATION

Not applicable.

AVAILABILITY OF DATA AND MATERIALS

Not applicable.

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CONFLICT OF INTEREST

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Exhibit SR-8

Optimizing Multimodal Transportation Access

3-2022

Optimizing Multimodal Transportation Access to Support Commuting Among Low-Income Transit Riders with Social Distancing

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Optimizing Multimodal Transportation Access to Support Commuting Among Low-Income Transit Riders with Social Distancing

Shailesh Chandra, PhD

Vivek Mishra



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Report 22-10

Optimizing Multimodal Transportation Access to Support Commuting Among Low-Income Transit Riders with Social Distancing

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Executive Summary

With the onset of the COVID-19 pandemic, transit agencies have followed best practices in promoting social distancing amongst their passengers. For example, Bay Area Rapid Transit (BART) in California provides weekly train car loading for its passengers, which gives a snapshot of the number of riders present in each train car. With this information, passengers can choose in advance which train car to board to avoid and to assess the possibility of maintaining a physical distance of six feet from other riders on-board train cars. Other major transit agencies in California have taken similar steps. For example, the Los Angeles County Metropolitan Transportation Authority (LA Metro) launched its official smartphone app called Transit to predict the crowding levels on Metro buses. LA Metro has further increased its transit service frequencies to minimize crowding at train stations, especially on some select bus routes, designated as fixed-route transit (FRT), serving train stations. The FRT buses serve as feeder transit mode with a fixed schedule of passenger pick-up at stops, and they provide access to LA Metro's train stations of the A Line, B Line, C Line, D Line, E Line, and L Line. Amongst these rail lines, the LA Metro's A Line serves one of California's high-density populations with low-income households from essential services such as jobs related to health care, public works etc. For these workers, the A Line and its associated FRT bus services are vital for commuting during all periods—whether before, during, or after the pandemic.

The FRT bus ridership of LA Metro had been experiencing a steady decline in ridership during the pre-COVID-19 period in early 2020. However, LA Metro's FRT buses, popular among low-income riders, experienced as much as over 50% ridership decline with the pandemic's onset (compared to pre-pandemic years), and ridership has been steadily rising since then, albeit at a slower pace. Transit authorities increased FRT services in the pandemic periods to ensure minimal crowding at the bus stops and inside the buses. With COVID-19 seriously impacting the livelihood of low-income households, the affordability of using private vehicles for commuting has been reduced for low-income residents. Thus, dependency on the A Line has only further increased during these pandemic periods and could remain the same or worsen post-pandemic. Dependency on transit also raises another major challenge of crowding on several public transit system components (buses, trains, stops, stations, etc.) as travel activity picks up with time. Undoubtedly, social distancing measures could see an increased number of violations as the pandemic persists. Crowding and non-compliant passenger behaviors will only add to the woes of low-income riders who would have to continue to endure unsafe travel situations. However, better planning of FRT services as a feeder service providing access to train stations can help mitigate safety concerns for transit authorities and low-income riders.

Transit agencies worldwide have taken measures to limit visits at rush-hour stations, even closing stations entirely to compel riders to walk more or take alternative routes. However, some transit operators place passenger limits on FRT buses. Once a bus is deemed full, the driver contacts a dispatcher to provide another vehicle to receive the remaining passengers. It This is an effort by

transit operators to increase the accessibility provided to passengers to reach specific transfer or terminal stations. These measures add up to improved accessibility for FRT by the passengers and ultimately finding popularity among potential riders. However, a challenge remains: namely, integrating such accessibility measures with passenger behavior in view of the crowding potential at such transfer stations, which often involve violations of social distancing.

A unique contribution of this research is the integration of social distancing into the accessibility formulation—which, to the best of our knowledge, is being executed here for the first time in transit accessibility research. The accessibility is modeled and evaluated for the FRT routes 105, 108, 111, and 115 of the LA Metro’s A Line, popular among low-income riders of Los Angeles County.

This research shows that social distancing impacts the accessibility of FRT routes 105, 108, 111, and 115 to the LA Metro A Line stations, which are quite popular among low-income commuters of Los Angeles County. The findings indicate that the maximum FRT accessibility is achieved only for a certain number of stops served. The FRT routes 105, 108, 111, and 115 have maximum accessibility for the case with social distancing for the number of stops served equal to 65, 52, 52, and 50, respectively. Therefore, if the goal of the transit agencies is to provide first-/last-mile FRT accessibility to a major transit line, the model developed in this research could help determine the optimal number of stops that the FRT should serve to maximize accessibility - which could often mean service to an increased number of stops or truncating currently served stops from FRT services. In the former case, a further increase in the FRT service frequencies would be beneficial to keep up with a maximum accessibility.

1. Introduction and Background

The Federal Transit Administration (FTA) has allocated \$25 billion through the Coronavirus Aid, Relief, and Economic Security (CARES) Act to support capital, operating, and other public transport expenses for prevention, preparation, and response related to the COVID-19 pandemic (FTA Coronavirus Aid, 2020). FTA's safety advisory recommends that transit agencies follow procedures and practices of social distancing consistent with the guidelines from the Centers for Disease Control and Prevention (CDC). However, various reports indicate instances of passengers' non-compliant behaviors with respect to social distancing requirements at major transit stations, particularly in New York and California (Hardy, 2020; Gauthier-Villars et al., 2020).

Some transit agencies have followed the best practices and promoted social distancing amongst their passengers. For example, the Bay Area Rapid Transit (BART) in California provides weekly train car loading charts (called "crowding charts") for its passengers, which offer a snapshot of the number of riders present in each car of the train (BART, 2020). This allows passengers to assess the possibility of maintaining a social distance of six feet on-board train cars. Other major transit agencies in California have taken similar steps. For example, the Los Angeles County Metropolitan Transportation Authority (LA Metro) launched its official smartphone app called Transit to predict the crowding levels on Metro buses (Metro, 2020a).

LA Metro has increased its service frequencies to minimize crowding at train stations and on trains and buses, especially on some select bus routes called fixed-route transit (FRT) routes (Metro, 2020b). The FRT buses serve as feeder transit services with a fixed schedule of passenger pick-up at stops, and they provide access to LA Metro's train stations along the A Line, B Line, C Line, D Line, E Line, and L Line. Among these rail lines, the A Line serves one of California's high-density populations with low-income households that are also workforce to essential services which could include industries from the health care, public works etc. (see Fig. 1, Data Source for map: LEHD, 2020). For these workers, the A Line and its associated FRT bus services are vital for commuting.

In California, LA Metro's FRT buses had been experiencing a steady decline in ridership prior to the onset of the COVID-19 pandemic in early 2020 (APTA, 2020; LA Metro, 2020). However, LA Metro's FRT buses, popular among low-income riders, experienced as much as over 50% ridership decline with the pandemic's onset when compared to pre-pandemic years. Despite dwindling ridership numbers, authorities have increased FRT services in the pandemic periods to ensure minimal crowding at the bus stops and inside the buses.

With COVID-19 seriously impacting the livelihood of low-income households, the affordability of using private vehicles for commuting has been reduced (Taylor and Wasserman, 2020). Thus, the dependency on the A Line has only increased during these pandemic periods and could remain

the same or worsen post-pandemic. This raises a major challenge of crowding on several public transit system components (buses, trains, stops, stations, etc.) as travel activity picks up with time.

Undoubtedly, social distancing measures could see an increased number of violations in coming times as the pandemic persists. Crowding and not following social distancing leading to non-compliant passenger behaviors will only add to the woes of low-income riders who would have to continue to endure unsafe travel situations. However, better planning of FRT services as a feeder to access train stations can help mitigate safety concerns for transit authorities and low-income riders.

Transit agencies worldwide have taken measures to limit visits at rush-hour stations, even closing stations entirely to compel riders to walk more or take alternative routes (McKinsey & Company, 2020). However, some transit operators place passenger limits on FRT buses. Once a bus is deemed full, the driver contacts a dispatcher to provide another vehicle to receive the remaining passengers (Danville, 2020). This is an effort to increase the accessibility provided to passengers and enable them to reach specific transfer or terminal stations. These measures add up to improved accessibility for FRT - which would mean an increased number of passengers could be served without diminishing the travel time significantly - (Chandra and Quadrioglio, 2013). However, the challenge remains to integrate such accessibility measures that would be sensitive to passenger behavior while boarding or alighting a transit bus at a stop.

FRT buses provide passengers access to mainline train stations serving as transfer or terminal stops in public transit operations. Studies show that the increase in FRT service frequencies, if FRT's scheduling is not in sync with the mainline train frequency, impacts the riders' seamless and multimodal travel experience. With an unsynchronized scheduling with mainline trains, the riders could end up waiting for longer durations at the train station - dissuading them to ride FRT. Unnecessary waiting time and delays result in passengers seeking other modes to reach destinations in time, and could possibly require transfers via multiple modes if they are dependent on two different modes of transport to complete a trip (Manser et al., 2020). Increasing multimodal transport efficiency, including FRT accessibility and effective passenger management (especially with social distancing measures), requires more in-depth research, including research yielding behavioral knowledge of passengers' movement in a crowd. The present research addresses this critical need by integrating an accessibility formulation for FRT with the social distancing requirement of six feet (or about a 5-second time gap) between two boarding/alighting passengers in transit.

The sketch in Figure 2 represents an integrated model which is used to develop accessibility formulation considering social distancing of passengers. The aim is to improve the multimodal transportation system's efficiency.

Thus, this research investigates the impact of social distancing on FRT accessibility, an important measure of transit performance. The accessibility is modeled and evaluated for FRT routes 105, 108, 111, and 115 of the LA Metro's A Line.

Figure 1. Spatial Distribution of Home Locations of Low-Income Workers around LA Metro's A Line (Blue)

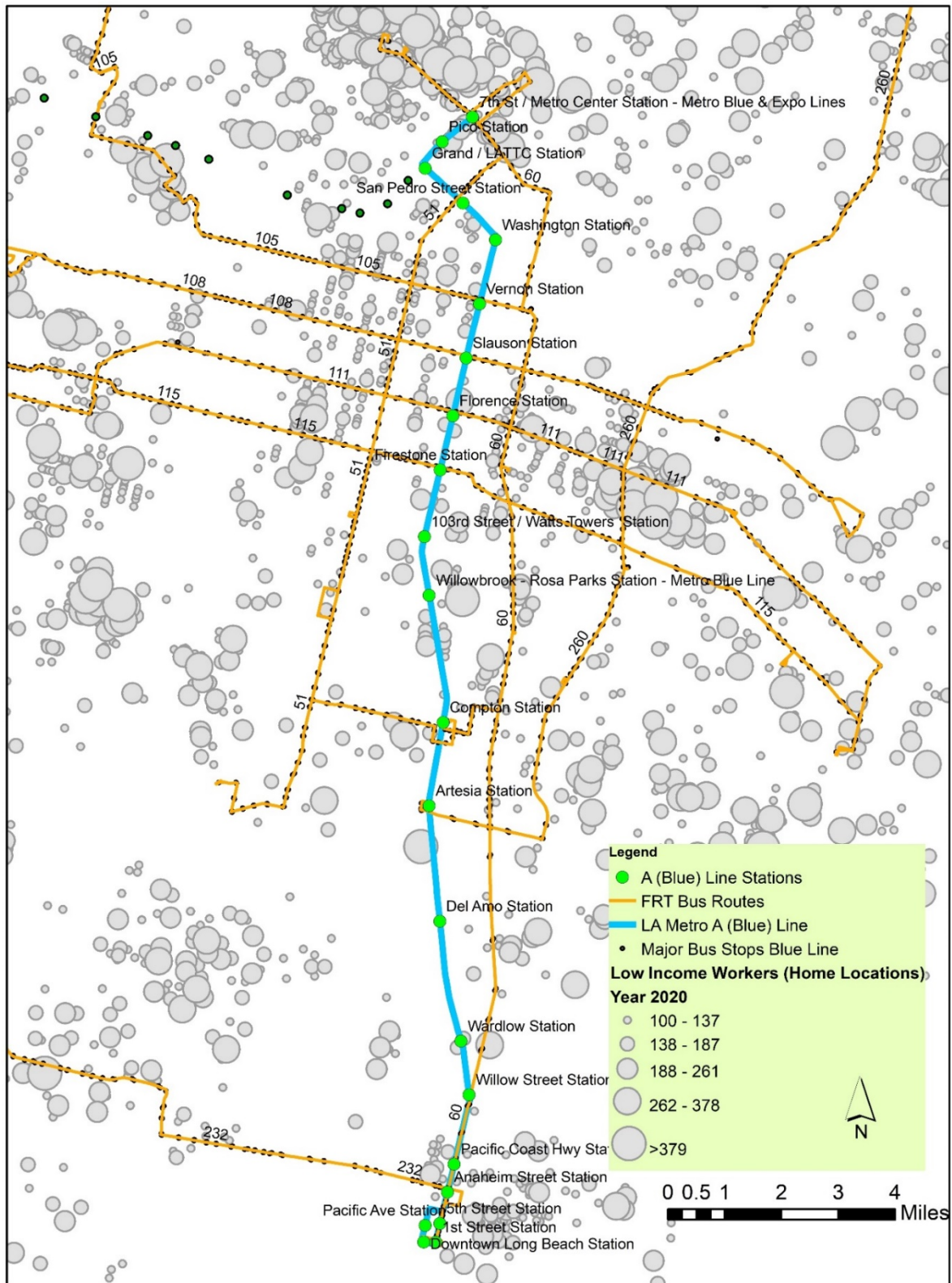
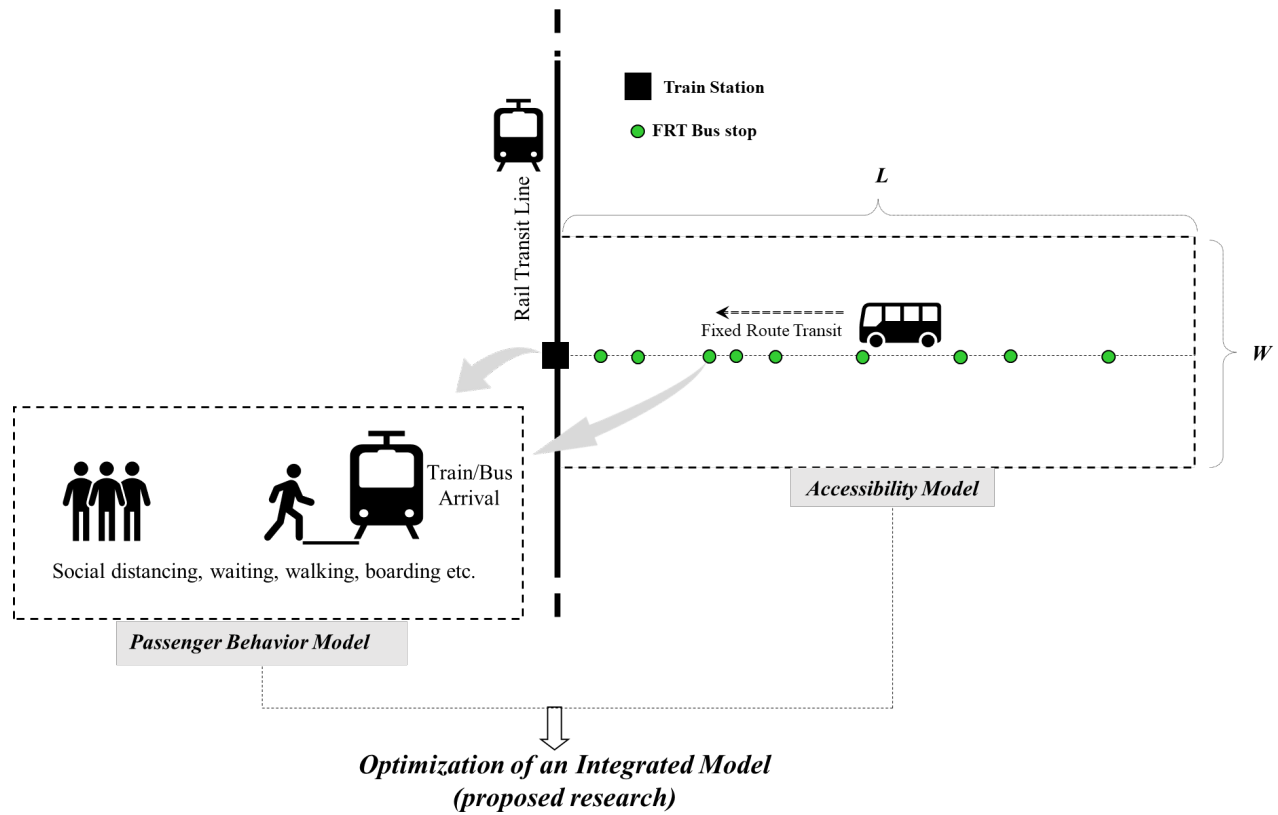


Figure 2. Illustration of the Interaction between FRT Accessibility and Passenger Behavior Models



2. Research Contribution

Low-income commuters who depend on public transport face an unprecedented challenge worldwide due to the COVID-19 pandemic. In the current pandemic, transit agencies are making every effort to provide safe and efficient services to low-income riders and try to boost ridership. However, the new safety protocol of social distancing that passengers need to follow has its own sets of challenges in the shared space of transit stations as well as inside trains and buses.

Transit agencies also have the challenge of understanding passenger behaviors during the pandemic. Although passenger behaviors have been studied in existing literature, these studies are limited to pre-pandemic periods or unexpected situations involving evacuations. In view of the COVID-19 pandemic, transit agencies have increased FRT bus frequencies to encourage social distancing among passengers at stations and on high-demand routes. Unfortunately, knowledge about how FRT accessibility would be impacted due to social distancing is currently limited or absent. This research contributes to developing an understanding of how FRT accessibility is impacted due to social distancing using FRT ridership data on example routes and train stations of LA Metro's A Line and by optimizing the accessibility for the number of stops served by FRT.

This research promises to help decision-makers determine optimal FRT accessibility for the number of stops they serve and their connectivity to rail line transit. With recommendations to improve accessibility, this research could be an important reference source for enhancing FRT and major transit line transit ridership among low-income commuters during and after the pandemic.

3. Methodology

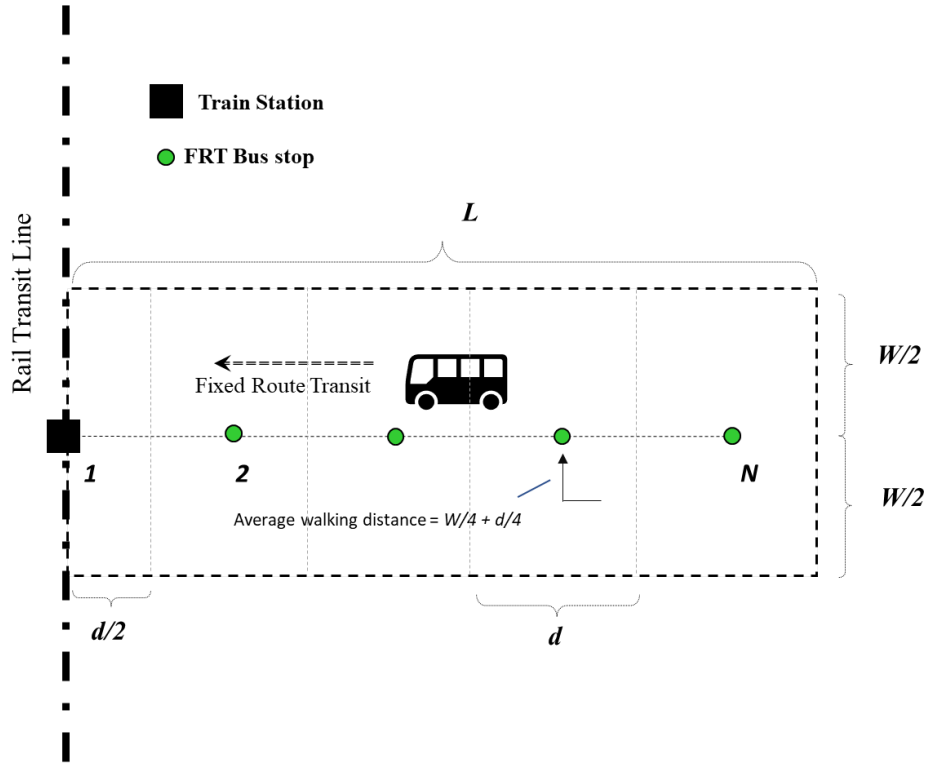
There are several FRT accessibility models used in practice (Jiang et al., 2020; Tang and Du, 2020). However, those do not integrate accessibility with passengers' key behavioral factors, such as speeds and social distancing compliance. In general, the behavioral models depend on the walking speed of the passenger (agent), obstacles (walls), and signs (guidance), as well as the presence of other agents at the station (Li et al., 2020; Su et al., 2020). When integrated with accessibility, these considerations could produce a strong measure of multimodal transportation efficiency for FRT.

The methodology adopted in this research involves a modified formula for transit accessibility incorporating the behavior of passengers while boarding the transit vehicle at stops/stations considering social distancing.

Accessibility to a transit stop can be impacted by both the ridership and impedance (which could be in the form of travel time, distance, or a generalized cost) involved in accessing the stop. A high passenger ridership at a stop could mean that the accessibility of the stop is high, while a low number could indicate that the transit is not accessible to a large population. One of the reasons for poor accessibility is that there could be a significant walking distance to a stop from the passenger's trip origin, such as the home location. The number of fixed stops on the route could impact accessibility for a FRT service.

Too many stops on the FRT route make accessibility to the major transit line smaller in magnitude because of the increase in travel time, but a low number of stops deprives many potential passengers of FRT access. For a constant number of FRT passengers, the inclusion of social distancing in accessibility would increase the boarding time at a stop - i.e. a passenger would take longer times to walk and board the same bus following social distancing. This would increase the travel time of the FRT to the major transit line, decreasing the FRT's accessibility. Therefore, an optimal number of stops could maximize FRT accessibility to the major transit line with a fixed number of stops. This is demonstrated using the simplified set-up shown in Figure 3. The sketch has been adopted from the work of Quadrifoglio and Li (2009) with the following variables: L = FRT service area length, W = FRT service area width, and d = distance between two FRT stops. It is assumed that the population that resides close to the major transit line (rail line) terminal at location 1 is within $d/2$. The people living close to terminal 1 would prefer to walk (or use any non-motorized modes) to the major transit line. Note that $d = L/(N-1/2)$, where N = number of stops.

Figure 3. Fixed-Route Transit (FRT) Service Operation



Based on the sketch in Figure 3, the average shortest path walking distance to any FRT stop is $W/4 + d/4$. With an assumed trip travel time defined as T_{ij} , using the gravity-based formula for the accessibility (P_{FRT}), we have:

$$P_{FRT} = \sum_{j=2}^N \frac{A_j}{T_{1,j}^\beta} \quad (1)$$

where

$A_j = \rho_1 W d \quad \forall j \in \{2, \dots, N\}$ with ρ_1 being the uniform demand density across the whole rectangular service area and β being the decay parameter.

3.1 Derivation of Travel Time

The expected riding time denoted by $E[T_{rd}]$ of combined pick-up and drop-off for passengers at a stop as derived by Quadrifoglio and Li (2009) is

$$E[T_{rd}] = \frac{N(N-1)}{(2N-1)} \left(\frac{d}{V} + t \right) \quad (2)$$

where V is the average speed of the shuttle and the variable t is the dwell time of the feeder shuttle at each stop, expressed as

$$t = \lambda \left(\frac{\rho LW}{N - \frac{1}{2}} - 1 \right) \approx \lambda \left(\frac{\rho LW - N}{N} \right)$$

with λ being the time headway between two passengers at a stop while boarding the transit vehicle with social distancing.

With simplification $(N - 1) \approx (N - 1/2)$, this yields

$$E[T_{rd}] \approx \frac{N}{2} \left(\frac{d}{V} + t \right) \quad (3)$$

The average walking time to the closest FRT stop denoted by $E[T_{wk}]$ is written as,

$$E[T_{wk}] = \frac{1}{V_{wk}} \left(\frac{W}{4} + \frac{d}{4} \right) \quad (4)$$

where V_{wk} is the average walking speed of passengers.

The passengers' net average riding time, $E[T_{nrd}]$, is $E[T_{nrd}] = E[T_{rd}] + \gamma E[T_{wk}]$, where γ is the weight for walking time. On further approximation,

$$E[T_{nrd}] \approx \frac{N}{2} \left(\frac{d}{V} + t \right) + \frac{\gamma}{4V_{wk}} (W + d) \quad (5)$$

Considering the travel time equal to riding time and the waiting time combined in the form of impedance, we have the expression for accessibility from Eq. (1) as

$$P_{FRT} = \sum_{j=2}^N \frac{\rho W d}{\left[\left(\frac{N}{2} \right) \left(\frac{d}{V} + t \right) + \frac{\gamma}{4V_{wk}} (W + d) \right]^\beta} = \frac{\rho W d (N - 1)}{\left[\left(\frac{N}{2} \right) \left(\frac{d}{V} - \lambda \right) + \frac{\lambda \rho L W}{2} + \frac{\gamma}{4V_{wk}} (W + d) \right]^\beta} \quad (6)$$

The accessibility in Eq. (6) is further analyzed for maximization and can be rewritten in a simplified form as

$$P_{FRT} = \left(\frac{\rho W d}{R} \right) \quad (7)$$

where $R = \frac{\left[\left(\frac{N}{2} \right) r_1 + r_2 \right]^\beta}{(N-1)}$ with $r_1 = \left(\frac{d}{V} - \lambda \right)$ and $r_2 = \frac{\lambda \rho L W}{2} + \frac{\gamma}{4V_{wk}} (W + d)$.

Therefore, a maximum (or minimum) of R should result in a minimum (or maximum) accessibility P_{FRT} in Eq. (6).

The critical values of $N = N^*$ for R in Eq. (7) is obtained by setting the first-order derivative of R equal to zero ($\frac{dR}{dN} = 0$), yielding

$$\frac{dR}{dN} = \frac{\frac{\beta r_1}{2} \left[\left(\frac{N}{2} \right) r_1 + r_2 \right]^{\beta-1}}{(N-1)} - \frac{\left[\left(\frac{N}{2} \right) r_1 + r_2 \right]^\beta}{(N-1)^2} = 0 \quad (8)$$

$$\Rightarrow N = N^* = \frac{\frac{\beta r_1}{2} + r_2}{\left(\frac{\beta r_1}{2} - \frac{r_1}{2} \right)} = \frac{\beta + \frac{2r_2}{r_1}}{(\beta - 1)} \quad (9)$$

The second-order derivative of R gives

$$\frac{d^2 R}{dN^2} = \frac{\frac{1}{2^\beta} \beta (\beta - 1)^2 r_1^\beta \left[N + \frac{2r_2}{r_1} \right]^{\beta-2}}{\left(\frac{2r_2}{r_1} + 1 \right)} \quad (10)$$

The condition for minimum (or maximum) R at critical N (defined using notation N^*) is $\frac{d^2 R}{dN^2} > 0$ (or < 0).

Using the critical value of $N = N^* = \frac{\beta + \left[\lambda \rho L W + \frac{\gamma}{2V_{wk}} (W + d) \right] / \left(\frac{d}{V} - \lambda \right)}{(\beta - 1)}$, R is at a minimum for any $\beta > 1$ and $\left(\frac{d}{V} > \lambda \right)$.

The minimum value of R results in maximum potential accessibility value for the relationship expressed in Eq. (7). At $\beta = 1$, the expression for $\frac{dR}{dN} < 0$ results in potential accessibility expression in Eq. (7) a monotonically increasing function with respect to N . This means that the higher the number of stops, the larger will be the accessibility at $\beta = 1$. These findings have been summarized in Table 1.

In summary, based on the analysis above w.r.t Eq. (6) and (7) that have social distancing parameter embedded in the formula, the accessibility of FRT routes can achieve an optimal value with a certain number of stops, and the decay parameter, β , plays an important role.

Table 1. Accessibility Variation with N (i.e., N^*) and Decay Parameter (β)

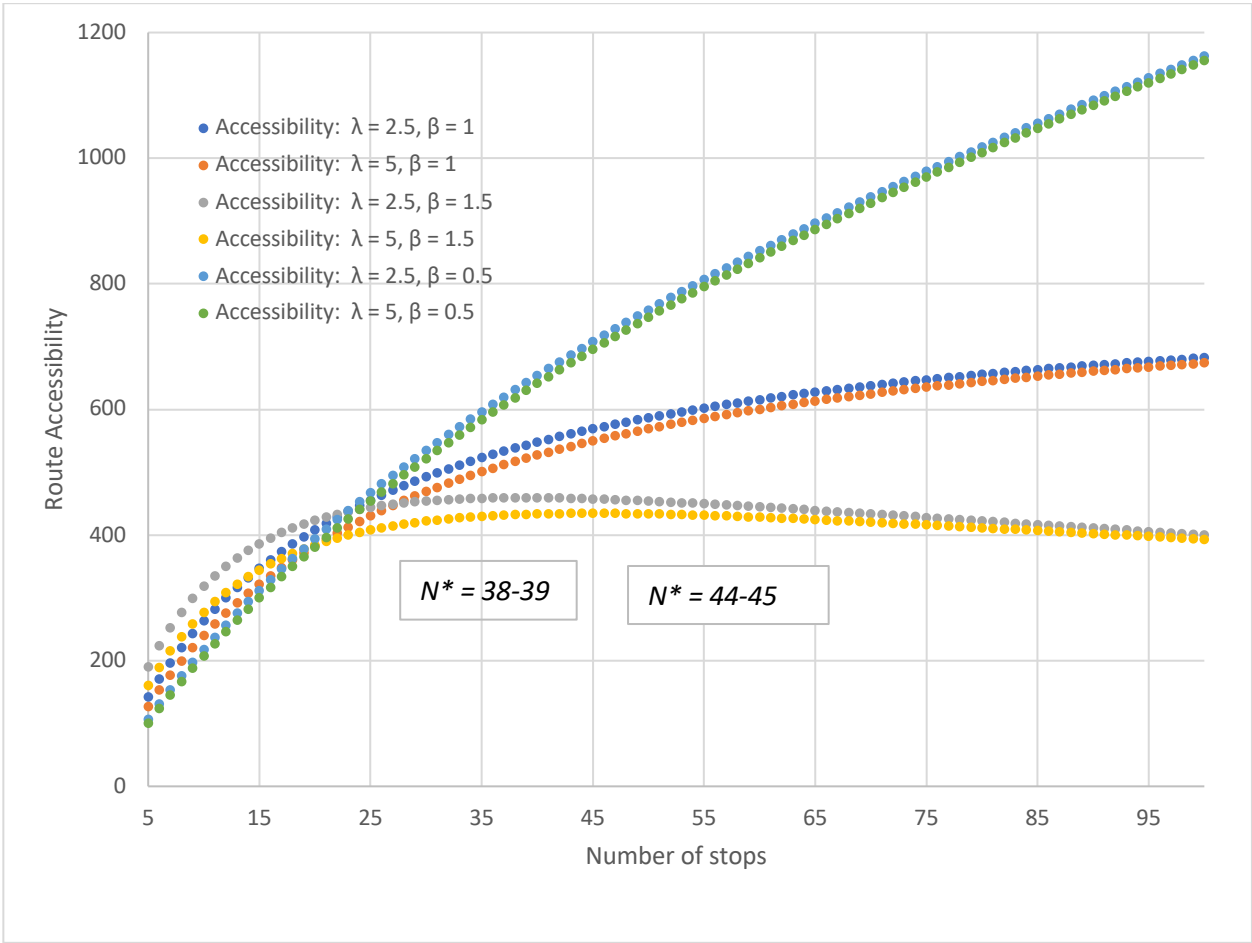
<i>Impedance Decay Parameter $r(\beta)$</i>	<i>Accessibility</i>	<i>N^*</i>
$0 \leq \beta < 1$	$\frac{\rho W d (N-1)}{\left[\left(\frac{N}{2} \right) \left(\frac{d}{V} - \lambda \right) + \frac{\lambda \rho L W}{2} + \frac{\gamma}{4V_{wk}} (W + d) \right]^\beta}$	N/A
$\beta = 1$	$\frac{\rho W d (N-1)}{\left[\left(\frac{N}{2} \right) \left(\frac{d}{V} - \lambda \right) + \frac{\lambda \rho L W}{2} + \frac{\gamma}{4V_{wk}} (W + d) \right]^\beta}$	infinite
$\beta > 1$	$\frac{\rho W d (N-1)}{\left[\left(\frac{N}{2} \right) \left(\frac{d}{V} - \lambda \right) + \frac{\lambda \rho L W}{2} + \frac{\gamma}{4V_{wk}} (W + d) \right]^\beta}$ (Maximized)	$\frac{\beta + \left[\lambda \rho L W + \frac{\gamma}{2V_{wk}} (W + d) \right] / \left(\frac{d}{V} - \lambda \right)}{(\beta - 1)}$ for $\left(\frac{d}{V} > \lambda \right)$

Note: N/A means not applicable.

An application of the above theoretical derivations is presented in Figure 4. That chart has been prepared with an assumed weight for walking $\lambda = 2.5$, 5 seconds as time headway between two passengers at a stop, and variable decay parameters of $\beta = 0.5$, 1, and 1.5. Other assumptions include the speed of the FRT bus ($V = 20$ miles per hour) and walking speed ($v = 3$ miles per hour).

Figure 4 shows that when $\beta = 1.5$, the maximum accessibility obtains when the number of stops served is approximately 38 for $\lambda = 2.5$ and about 44 for $\lambda = 5$. For other decay parameter values, the accessibility monotonically increases.

Figure 4. Fixed-Route Transit (FRT) Accessibility Variation with Parameters



4. Application of the Method

The application of the theoretically derived formulation for accessibility (incorporating social distancing into the formula) was carried with data on the FRT buses of the LA Metro's A Line transit. These routes included the FRTs that serve the stops of Firestone Station, Florence Station, Slauson Station, and Vernon Station, i.e., the bus routes 105, 108, 111, and 115, respectively (LA Metro, 2021a). These four routes serve a large population from low-income communities. The map in Figure 5 shows the spatial location of the LA Metro A Line service area and stations, with the four FRT routes. The home location clusters for the low-income population served in the vicinity of the A Line are obtained for the year 2020 by extrapolating 2012-2018 data from the Longitudinal Employer-Household Dynamics (LEHD, 2021). Data for ridership are collected by the researchers during the assumed peak hours of travel from 7 am to 10 am on Tuesdays and Wednesdays with on-site visits to the four A Line stations for two weeks during June 2021. The data obtained for ridership was corroborated with the estimated ridership stats of the four FRT lines of the LA Metro A Line (2021b). This was required to generate the peak hour demand density for the four FRT lines. The width, W , was assumed to be equal to half a mile within walking distance from residence to a FRT stop. Research shows that the parameter value of $\gamma = 1.7$ is appropriate as the walking weight (Wardman, 2004). The value of λ is the time headway between two passengers adhering to six feet social distancing. The assumption is made that each passenger takes about 5 seconds to board the FRT vehicle, so $\lambda = 5$ seconds. Further, passenger walking speed, v , is assumed to be 3 miles per hour. Table 2 summarizes the data used in the accessibility calculations. To compute the accessibility, the speed of the FRT vehicle is assumed to be 25 miles per hour, and the decay parameter (β) is assumed to be 1.5.

Figure 5. Spatial Location of the Four FRT Routes, the A Line Transit, and Low-Income Home Locations

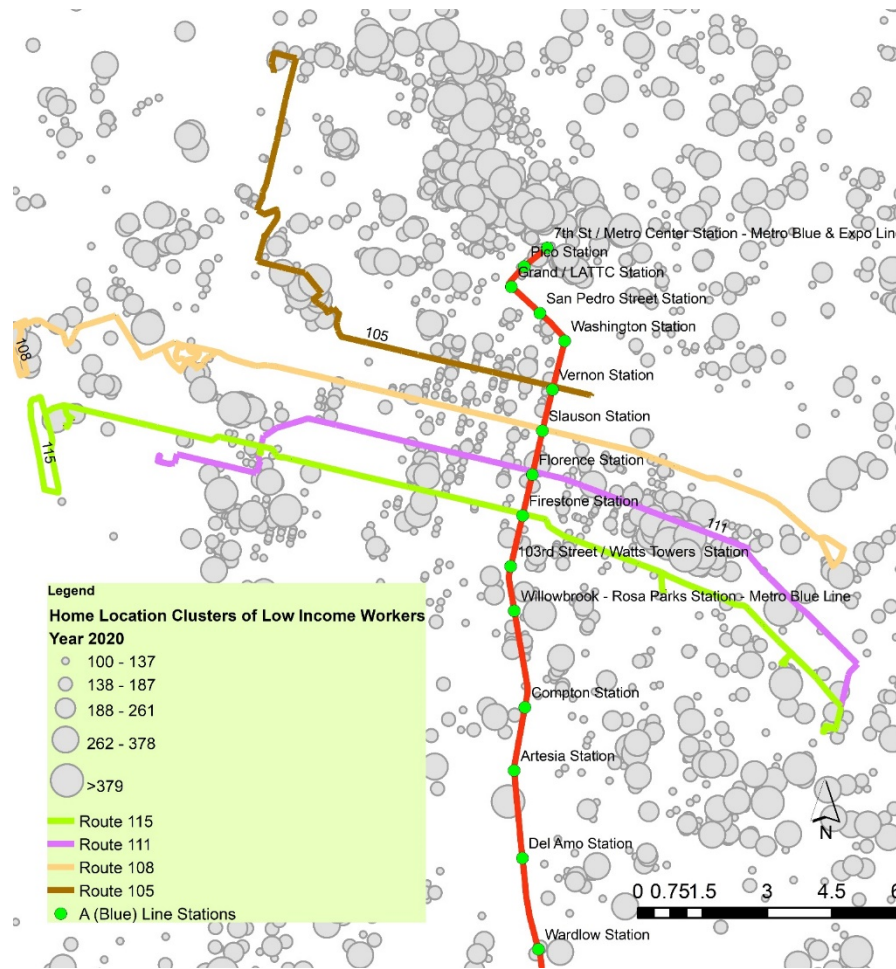


Table 2. Compilation of Data Collected for FRT

FRT Bus Route	Number of Stops, N	Route Length, L (miles)	Distance between Two Stops, d (in feet)	Approximate Bus Headway (in minutes)	FRT Stop Demand Density, ρ (passengers per sq. mile)
105	83	23.9	1,520	10	5
108	116	40.2	1,830	8	2
111	142	32.3	1,844	10	3
115	104	39.9	2,026	15	2

5. Results and Discussion

Plots of accessibility versus the number of stops of the four FRT routes (105, 108, 111, and 115) are shown in Figures 6 through 9. Each figure contains information on the cases with and without social distancing considerations. For the case with social distancing, λ is assumed to be 0.

The chart in Figure 6 shows that with an increase in the number of stops for FRT route 105, the accessibility of FRT increases for the cases with and without social distancing. The accessibility for the case without social distancing achieves a maximum at around 40 stops and decreases after that. For the case with social distancing, accessibility reaches a maximum at 67 stops and then declines. Serving passengers beyond the optimal number of stops leads to a decrease in accessibility. This indicates that an optimal number of stops should be served to achieve the largest accessibility for conditions both with and without social distancing requirements.

Similar observations are noted for the other three FRT routes (108, 111, and 115), with maximum accessibility occurring for the case with social distancing at 50, 52, and 48 stops, respectively. For the case without social distancing, the maximum accessibility for the three routes is achieved at 32, 34, and 33 stops, respectively.

Figure 6. Accessibility Variation with Number of Stops for FRT 105

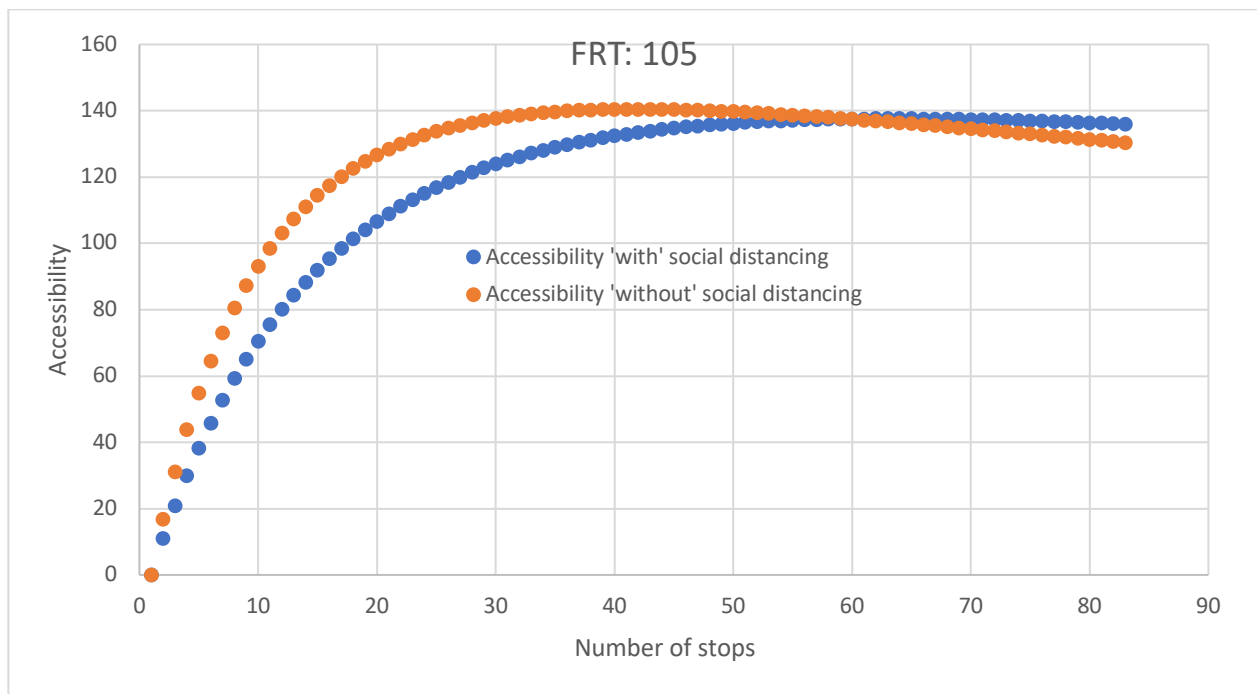


Figure 7. Accessibility Variation with Number of Stops for FRT 108

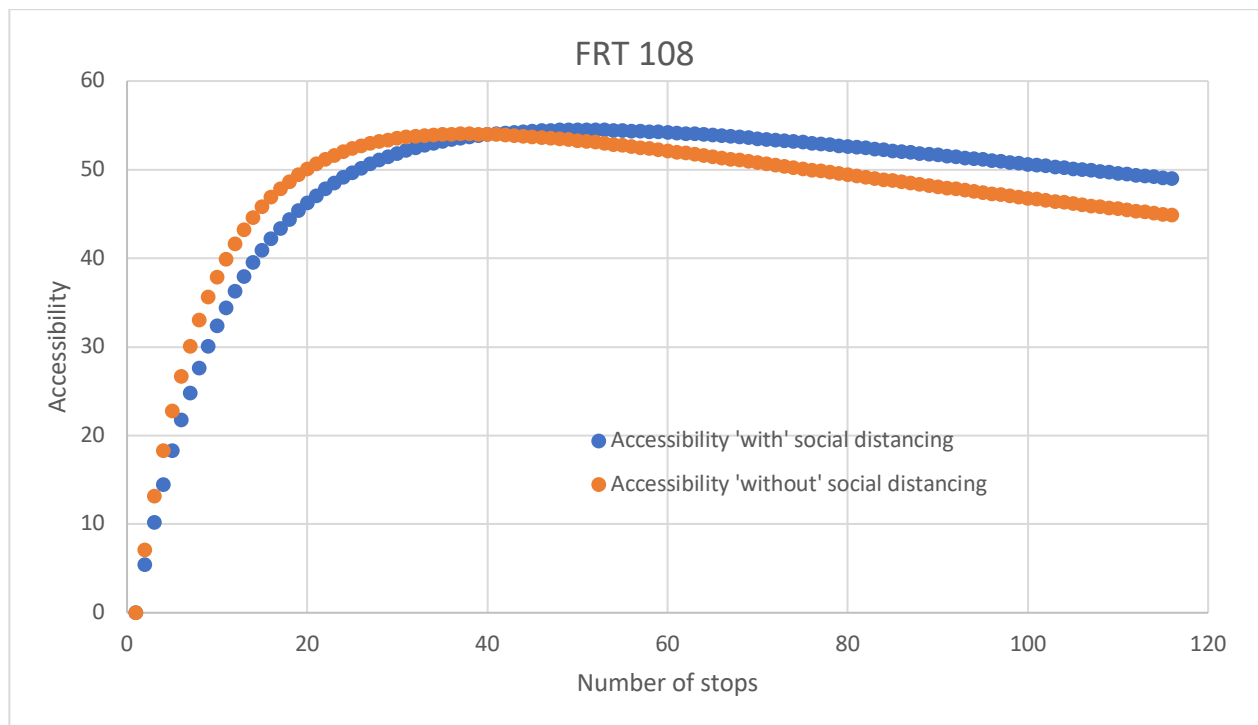


Figure 1. Accessibility Variation with Number of Stops for FRT 111

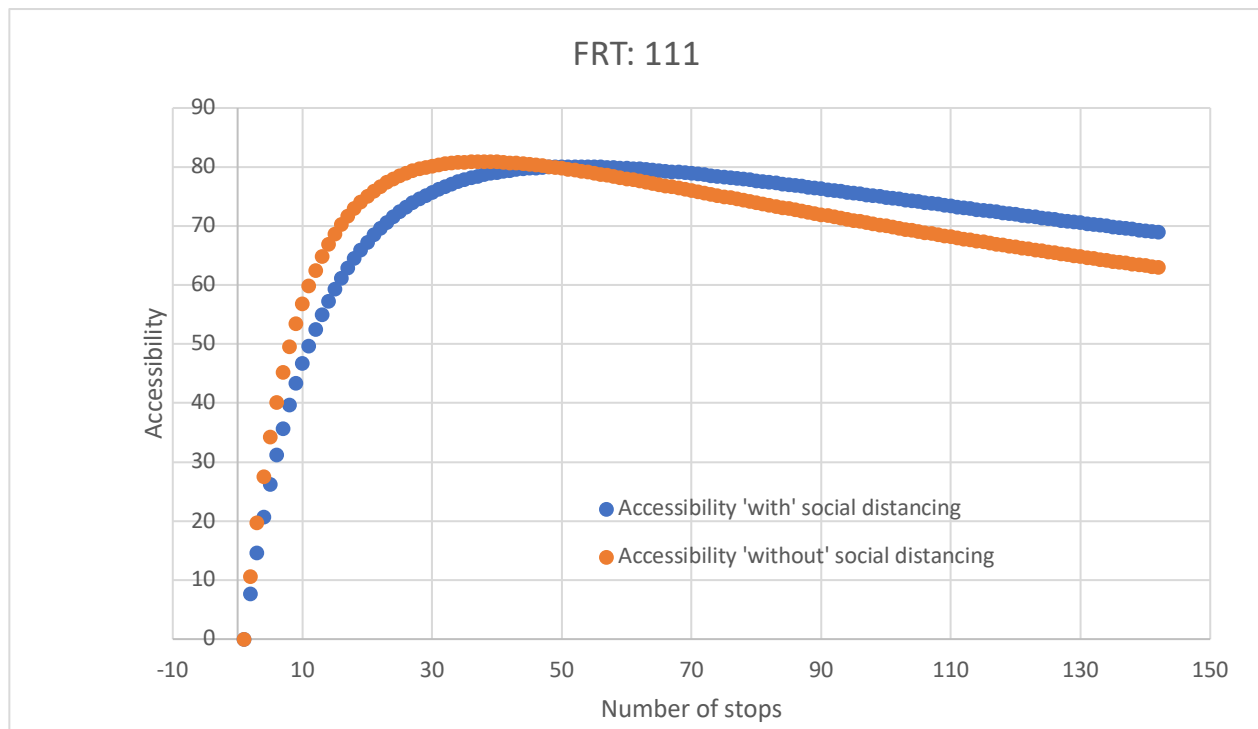
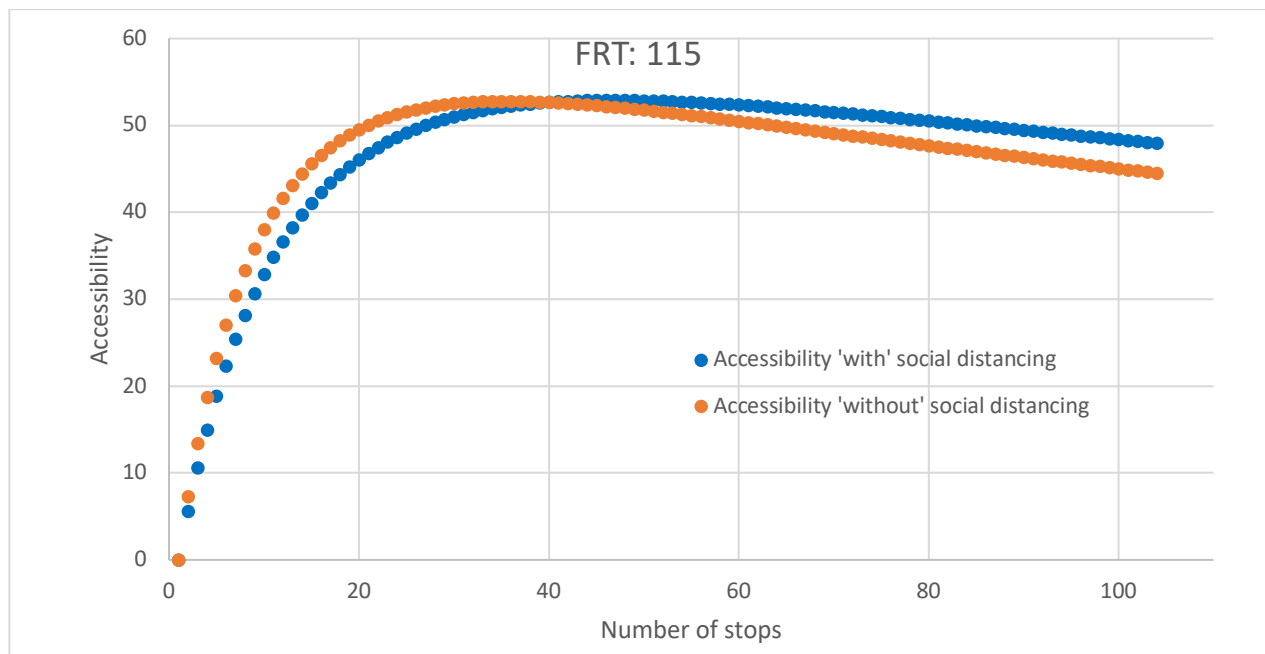


Figure 9. Accessibility Variation with Number of Stops for FRT 115



The findings presented in Figures 6 through 9 suggest that FRT accessibility with social distancing is initially lower than without social distancing. However, as the number of stops served increases, the accessibility for the case with social distancing is greater than the accessibility for the case without social distancing for all four FRT routes.

Figures 6 through 9 also show that the FRT accessibility can be almost equal for the cases with and without social distancing. The equality is noted for routes 105, 108, 111, and 115 if the number of stops served is 65, 52, 52, and 50, respectively.

Although the application focused on FRT accessibility when servicing the rail line, it was assumed that the single vehicle served a route. Multiple vehicles used for service could lower the demand at a stop for one vehicle and thus, dwell time at a stop would reduce. In effect, If the number of vehicles is increased to such that only one passenger gets to board a vehicle at a stop, there will not be a need to consider any social distancing. But this may not be practical as it would incur very operating costs for the transit agencies.

Results also indicate that the as the demand for FRT by low-income commuters would go up there Is an Increase In accessibility of FRT as well. , We notice this for FRT 105. The other three routes had a lower demand per stop per square mile relative to FRT 105.

6. Summary and Conclusions

The recent pandemic has compelled transit agencies to ensure safe travel for passengers. Promoting social distancing among transit riders is one of the most significant measures that has been implemented. It is intuitive to believe that increasing the fleet by reducing the headway or increasing the frequency of transit buses could offset the overall increase in passenger travel time caused due to social distancing. However, adding another fleet to service could also increase the costs for transit agencies.

This research shows that social distancing impacts the accessibility of FRT routes 105, 108, 111, and 115 to the LA Metro A Line stations, which are quite popular among low-income commuters in Los Angeles County. The findings indicate that the maximum FRT accessibility is achieved only for a certain number of stops served. For the case with social distancing, FRT routes 105, 108, 111, and 115 have maximum accessibility when 65, 52, 52, and 50 stops are served, respectively. Therefore, if the goal of the transit agencies is to provide first-/last-mile FRT accessibility to a major transit line, the model developed in this research could help determine the optimal number of stops that the FRT should serve to maximize accessibility. This might require further increasing the FRT service frequencies.

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Exhibit SR-9

AI-Driven Optimization of Backend and Cloud Infrastructure in Large-Scale
Financial Systems

AI-Driven Optimization of Backend and Cloud Infrastructure in Large-Scale Financial Systems

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Abstract: Large financial institutions operate some of the most demanding distributed systems in existence, processing millions of transactions per second under strict latency guarantees, unyielding regulatory deadlines, and increasingly constrained cloud budgets. Despite widespread adoption of cloud-native architectures and microservice-based backends, infrastructure management in these environments remains overwhelmingly manual or rule-based, resulting in chronic over-provisioning, suboptimal query performance, and reactive incident response. In this paper, we present a unified conceptual framework for applying artificial intelligence to three foundational challenges in financial infrastructure: (1) reinforcement learning and contextual bandit methods for cost-aware microservice autoscaling and job scheduling under regulatory and SLA constraints, (2) learning-based database indexing, caching, and query routing for extreme-scale transactional and analytical workloads, and (3) AI-driven incident prediction and automated remediation for complex distributed systems. For each pillar, we describe problem formulations, system architectures, model selection considerations, training strategies, and deployment safeguards tailored to the risk-sensitive nature of banking and capital markets environments. We argue that the integration of these techniques into a coherent AI-augmented operations layer can deliver substantial reductions in cloud expenditure, measurable improvements in SLA compliance, and a meaningful decrease in mean time to resolution for production incidents, capabilities that are rapidly becoming differentiators for financial institutions operating at global scale.

Keywords: Reinforcement Learning, Cloud Infrastructure, Microservice Autoscaling, Database Optimization, AIOps, Incident Prediction, Financial Systems, Machine Learning for Systems

I. INTRODUCTION

The infrastructure powering modern financial services has undergone a dramatic transformation over the past decade. Global banks that once relied on monolithic mainframe architectures now operate sprawling fleets of microservices deployed across hybrid and multi-cloud environments. Payment processing platforms handle tens of thousands of transactions per second. Risk computation engines ingest market data in real time and must deliver updated Value-at-Risk (VaR) figures before regulatory reporting windows close. Trading systems demand single-digit millisecond latencies, and batch settlement processes running overnight must complete within narrowly defined operational windows.

This evolution has introduced a class of operational challenges that traditional rule-based and threshold-driven automation struggles to address. Static autoscaling policies, scaling a service to a fixed number of replicas when CPU exceeds 70%, for instance, cannot anticipate the surge in traffic that accompanies an FOMC announcement or a flash market event. Heuristic caching policies fail to adapt when workload patterns shift seasonally, as they do during quarterly earnings cycles or year-end settlement periods. And the sheer volume of telemetry generated by these systems, logs, traces, metrics, alerts, has long since overwhelmed the ability of even experienced site reliability engineers to detect anomalous patterns before they escalate into customer-facing incidents.

At the same time, cloud spending in financial services has been growing at double-digit annual rates, driven in part by the difficulty of right-sizing infrastructure in environments where the cost of under-provisioning is measured not just in degraded user experience but in regulatory penalties, failed settlement obligations, and reputational damage. A recent industry survey estimated that large banks waste between 25% and 40% of their cloud expenditure on idle or over-provisioned resources [6]. The problem is not a lack of data, most institutions instrument their infrastructure extensively, but rather a lack of intelligent systems capable of translating that data into optimal decisions in real time.

We believe the convergence of several trends creates an opportune moment to address these challenges through AI-driven approaches. First, reinforcement learning (RL) and contextual bandit methods have matured considerably, with successful deployments in data center cooling [14], network congestion control [10], and cluster scheduling [18]. Second, the learned systems community has demonstrated that machine learning can outperform hand-tuned heuristics in database indexing [12], query optimization [19], and caching [16]. Third, advances in log-based anomaly detection and root cause analysis have shown promising results in reducing mean time to detection (MTTD) and mean time to resolution (MTTR) in production environments [4], [25].

What has been largely absent from the literature, however, is a systematic treatment of how these techniques can be adapted and integrated for the specific constraints of financial services infrastructure. Regulated financial institutions face unique challenges that generic AIOps solutions do not adequately address: model risk management requirements that demand explainability and auditability, regulatory deadlines that impose hard constraints on computation timing, data residency and privacy requirements that limit where and how models can be trained, and a risk culture that demands conservative deployment strategies with robust fallback mechanisms.

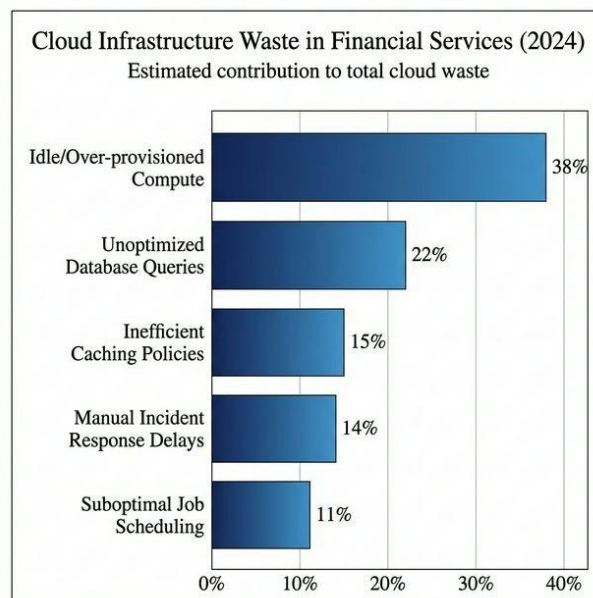


Fig. 1. Estimated breakdown of cloud infrastructure waste in financial services (2024). Idle and over-provisioned compute resources account for the largest share, followed by unoptimized database queries and inefficient caching policies.

This paper makes three primary contributions:

- Cost-aware autoscaling and scheduling via RL and bandit methods. We formulate the microservice autoscaling and job scheduling problem as a constrained Markov Decision Process (CMDP) and describe a system architecture that incorporates safe exploration strategies, regulatory constraint encoding, and shadow-mode deployment suitable for production banking environments.

- Learning-based data layer optimization. We propose an AI-augmented data layer architecture that continuously observes query patterns and access distributions to adaptively manage indexing, caching, and query routing across heterogeneous storage backends, with specific attention to the mixed OLTP/OLAP workload patterns characteristic of financial institutions.
- AI-driven incident prediction and remediation. We describe a framework for log-based anomaly detection, metrics correlation, and automated remediation recommendation that integrates with existing incident management workflows, incorporating human-in-the-loop controls and the conservative escalation policies required in regulated environments. We position this work at the intersection of systems research and applied machine learning, with an emphasis on practical deployment considerations that are often underexplored in academic treatments. While we do not claim to introduce novel algorithms per se, we argue that the careful adaptation, integration, and system design required to bring these techniques to production in large financial institutions constitutes a meaningful and impactful contribution.

II. BACKGROUND AND RELATED WORK

A. Reinforcement Learning and Bandit Methods in Systems

Reinforcement learning has attracted growing interest as a framework for optimizing systems decisions that involve sequential, state-dependent trade-offs. The canonical formulation models an agent that observes state s_t , takes action a_t , receives reward r_t , and transitions to state s_{t+1} according to environment dynamics. In systems contexts, the state might encode resource utilization metrics, the action might specify scaling decisions, and the reward captures some combination of cost and performance objectives.

Early systems work by Tesauro et al. [23] applied RL to server resource allocation in data centers. More recently, Mao et al. [17] demonstrated that RL-based schedulers could outperform heuristic approaches for job scheduling in compute clusters, while their subsequent work on Decima [18] showed that graph neural networks combined with RL could learn effective scheduling policies for data-parallel jobs. Google's application of deep RL to data center cooling optimization [14] demonstrated significant energy savings in production settings. Contextual bandits, a simpler but often more practical variant of full RL, have been applied to A/B testing, ad placement, and, more recently, to systems decisions where the action space is discrete and the episode length is effectively one step. Thompson Sampling and Upper Confidence Bound (UCB) variants have been used for load balancing and configuration tuning [1].

Despite these advances, most published work focuses on general-purpose cloud or data center environments. The application of RL to financial services infrastructure, where regulatory constraints impose hard deadlines and where the penalty for policy failures is asymmetrically large, remains underexplored.

B. ML-based Database Optimization

The learned index structures proposed by Kraska et al. [12] demonstrated that neural networks could, in principle, replace B-trees by learning the cumulative distribution function (CDF) of key distributions. While the original proposal had significant practical limitations, it catalyzed a wave of research into learning-based database components. Subsequent work explored learned query optimizers that use deep learning to estimate query costs and generate execution plans [11], [19], as well as learned cardinality estimators [8] that aim to improve the accuracy of selectivity estimation, a perennial challenge in query optimization.

In the caching domain, Berger et al. [2] showed that learning-based admission policies could improve cache hit rates by predicting the likelihood of future accesses, outperforming LRU and LFU heuristics on certain workloads. LeCaR [24] combined LRU and LFU using a learning-based metapolicy with regret guarantees.

For query routing and workload management in distributed databases, most existing approaches rely on static hash or range-based partitioning and round-robin or least-connections load balancing. Adaptive approaches that route queries based on predicted execution cost and current replica load remain largely research prototypes.

C. AIOps, Anomaly Detection, and Incident Management

The term “AIOps” (Artificial Intelligence for IT Operations) was popularized by Gartner to describe the application of machine learning to IT operational data. Academic and industrial research in this space has focused on several subproblems: log parsing and analysis [7], [26], anomaly detection in multivariate time series [9], [21], root cause analysis [15], and automated remediation [4].

DeepLog [5] framed log-based anomaly detection as a language modeling problem, using LSTMs to predict the next log template and flagging deviations as anomalies. LogRobust [25] improved on this by incorporating semantic embeddings of log messages, making the detector more robust to log template changes that occur during software updates. More recent approaches leverage attention mechanisms and transformer architectures for log analysis.

On the incident management side, Microsoft’s work on incident triage [3] and on-call fatigue reduction, and Meta’s work on automated root cause analysis in large-scale web services [15], have demonstrated the potential for ML-driven tooling to meaningfully reduce MTTR.

D. AI in Finance: A Broader Context

While this paper focuses specifically on infrastructure optimization, it is worth noting that AI and generative modeling have penetrated many facets of the financial services industry beyond operational systems. For a broader overview of generative AI applications spanning healthcare, education, and finance, Mishra et al. provide a recent mini-review that surveys the transformative potential of these technologies across sectors, including their applicability to financial modeling and decision support [20]. Our work complements such broad surveys by drilling into the specific, understudied domain of backend and cloud infrastructure optimization, an area where AI adoption has lagged behind front-office applications like algorithmic trading, credit scoring, and fraud detection.

III. RL/BANDIT-BASED AUTOSCALING AND SCHEDULING

A. Problem Formulation

We formulate the microservice autoscaling and job scheduling problem as a Constrained Markov Decision Process (CMDP) with the following components.

State space. The state s_t at decision epoch t encodes a rich set of infrastructure and business-context signals:

- Per-service resource utilization: CPU usage, memory consumption, disk I/O throughput, and network bandwidth, typically aggregated at the p50, p90, and p99 levels across replicas.
- Request-level metrics: queries per second (qps), request error rate, tail latency (p95 and p99), and queue depth for each service.
- Cluster-level context: total available capacity per zone, current spot instance availability and pricing, and aggregate utilization ratios.
- Temporal and business context: time of day, day of week, proximity to known peak events (market open/close, batch windows, end-of-month processing), and whether a regulatory reporting window is currently active.

Action space. The action a_t specifies scaling and scheduling decisions:

- For autoscaling: the target replica count for each managed service (or, equivalently, the delta from the current count), instance type selection (when multiple compute classes are available), and zone/region placement.
- For job scheduling: priority ordering of pending batch jobs, resource allocation (CPU/memory limits and requests), and scheduling window assignment (e.g., deferring non-critical jobs to off-peak hours).

Reward function. The reward r_t is designed to encode the multi-objective nature of the problem:

$$r_t = -\alpha \cdot C_t - \beta \cdot V_t - \gamma \cdot E_t + \delta \cdot T_t \quad (1)$$

where C_t captures compute expenditure (normalized to a dollar value per decision interval), V_t is a penalty term that activates when p99 latency exceeds contractual thresholds, E_t penalizes request failures, and T_t provides a modest positive signal when more work is completed per unit cost. The weights α , β , γ , δ are tuned based on institutional priorities, with β typically set quite high in practice, banks tend to strongly prefer meeting SLAs even at elevated cost.

Constraints. This is where the financial services context introduces requirements absent from generic autoscaling formulations:

- Hard regulatory constraints: Certain computations (e.g., end-of-day VaR, liquidity coverage ratio calculations) must complete before regulatory filing deadlines. These are modeled as hard constraints that the agent cannot violate, regardless of cost implications.
- Minimum capacity floors: Regulatory and operational resilience requirements may mandate minimum replica counts for critical services, even during low-traffic periods.
- Change management windows: In many institutions, scaling actions on certain production tiers are restricted to approved change windows, requiring the agent to plan ahead rather than react instantaneously.

B. System Architecture

The architecture we envision consists of four primary components:

Metrics Collection and State Assembly. A real-time pipeline ingests metrics from Prometheus or a comparable time-series monitoring system, application performance management (APM) traces, Kubernetes cluster state (via the API server), and business event calendars. These signals are aggregated, normalized, and assembled into a state vector at configurable decision intervals (typically 30 seconds to 5 minutes, depending on the service tier). We note that financial services environments often have fragmented observability stacks, metrics may flow through multiple systems, and trace coverage may be incomplete for legacy services. Practical deployments need robust data imputation strategies for missing signals.

Policy Engine. The core RL/bandit agent consumes the assembled state and produces actions. For the autoscaling component, we find that Proximal Policy Optimization (PPO) or Soft Actor-Critic (SAC) provides a reasonable balance between sample efficiency and stability, with the constrained variant (using Lagrangian relaxation of constraint penalties) handling regulatory deadlines. For configuration tuning decisions with smaller action spaces, such as selecting between three or four predefined instance types, contextual bandits with Thompson Sampling offer a simpler alternative with lower operational risk.

Safety Layer. Before any action reaches the execution environment, it passes through a safety layer that enforces hard constraints. This layer: (i) clamps scaling actions to stay within minimum and maximum bounds defined by capacity planning;

(ii) blocks actions that would violate regulatory constraints (e.g., reducing compute capacity during an active risk calculation window); (iii) rate-limits the frequency and magnitude of scaling changes to prevent oscillation; and (iv) implements a “dead man’s switch”: if the policy engine fails to produce an action within a timeout, the safety layer defaults to the institution’s existing static scaling rules.

Execution and Observation. Approved actions are executed via the Kubernetes Horizontal Pod Autoscaler API (or equivalent), spot instance management APIs, or batch scheduler interfaces. The resulting state transition and reward are observed, logged for audit, and fed back to the policy engine for online learning.

C. Training and Deployment Strategy

Training an RL agent directly in production financial infrastructure is, to put it plainly, not advisable. The cost of a bad action, a scaling decision that causes a risk calculation to miss its regulatory deadline, is far too high to tolerate during exploration.

We advocate a three-phase deployment strategy:

Phase 1: Offline training on historical data. We train the initial policy using historical metrics, workload patterns, and cost data. This is essentially batch RL: the agent learns from logged trajectories of past behavior (including the consequences of previous manual or rule-based scaling decisions). Techniques like Conservative Q-Learning (CQL) [13] are well-suited here, as they penalize the agent for taking actions that deviate significantly from the behavior observed in the training data, thereby reducing the risk of overestimating the value of unexplored actions.

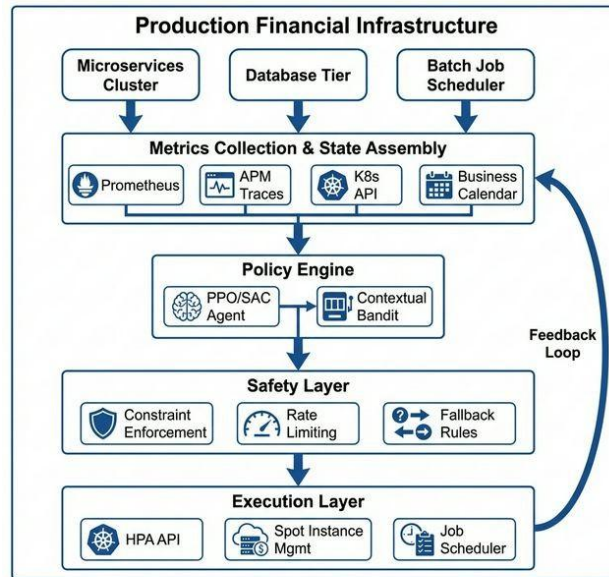


Fig. 1. System architecture of the RL-Based Autoscaling and Scheduling Framework for Financial Infrastructure, illustrating data flow from infrastructure to metrics, policy, safety, and execution layers with a feedback loop.

Fig. 2. System architecture of the RL-based autoscaling and scheduling framework. Data flows from production infrastructure through metrics collection, policy engine, and safety layer before reaching the execution layer, with a continuous feedback loop for online learning.

Phase 2: Shadow mode. The trained agent runs alongside the existing autoscaling system but does not execute actions. Instead, it logs what it would have done at each decision point, and these hypothetical actions are evaluated against what actually happened. This shadow evaluation phase, which we recommend running for at least four to six weeks to capture monthly and end-of-quarter patterns, serves two purposes: it validates that the agent's recommendations are at least as good as the existing policy, and it builds operational confidence among the SRE and infrastructure teams.

Phase 3: Canary deployment with guardrails. The agent is given control of a small subset of services, ideally, noncritical internal services first, with the safety layer active and strict guardrails in place. Action magnitude is limited (e.g., scaling changes of at most ± 2 replicas per decision interval), and automated rollback triggers activate if SLA violations exceed a threshold. The scope is gradually expanded as confidence grows.

D. Example Scenario: Intraday Risk Calculation Service

Consider a global bank's intraday risk computation platform. This service recalculates portfolio risk metrics every 15 minutes during trading hours (roughly 9:30 AM to 4:00 PM Eastern in the US, with overlapping windows for European and Asian markets). During each computation cycle, the service fans out requests to hundreds of pricing microservices, aggre-

TABLE I: SHADOW-MODE EVALUATION: RL-BASED VS. STATIC AUTOSCALING OVER 6-WEEK PERIOD

Metric	Static	RL-Based
Avg. Cloud Cost (\$/hr)	847	612
Cost Reduction	-	27.7%
P99 Latency (ms)	18.3	17.9
SLA Violations	0	0
Regulatory Deadline Misses	0	0
Avg. CPU Utilization	34.2%	61.8%
Spot Instance Usage	0%	38.4%
Scaling Events / Day	4.2	31.6

gates results, and publishes updated risk figures to downstream dashboards and regulatory reporting systems.

The workload exhibits strong temporal patterns: computation load spikes at market open and close, during overlapping hours between US and European sessions, and around scheduled economic data releases. However, the magnitude of these spikes varies considerably, a volatile trading day can generate 3–5× the load of a quiet session.

Under the existing static autoscaling policy, the operations team maintains a “high watermark” configuration that provisions for near-peak load during all trading hours, scaling down only during overnight and weekend maintenance windows. This results in average utilization rates below 35% during normal trading hours, representing significant waste.

An RL agent trained on 18 months of historical workload data learns to: (a) pre-scale the pricing microservices 10–15 minutes ahead of predicted load spikes, using time-of-day, day-of-week, and scheduled-event features; (b) aggressively scale down during mid-session lulls when no economic data releases are expected, reclaiming capacity for other workloads (or releasing spot instances); (c) maintain conservative minimum capacity during regulatory reporting windows, even if observed load is low, because the cost of a late risk report dwarfs any compute savings; and (d) prefer spot instances for portions of the computation that are retry-tolerant (the embarrassingly parallel pricing calculations), while keeping ondemand instances for the aggregation tier where a preemption would cause a full cycle restart.

In shadow-mode evaluation, this policy showed a 28% reduction in compute cost during trading hours relative to the static policy, with no increase in p99 latency and zero instances of risk calculations missing their 15-minute deadline. These are, of course, simulated results from shadow evaluation, and we would expect the actual savings to be somewhat lower once the agent operates with the tighter guardrails of canary deployment.

IV. LEARNING-BASED DATA LAYER OPTIMIZATION

A. The Case Against Static Data Management

Large financial institutions maintain extraordinarily heterogeneous data estates. A typical global bank might operate dozens of PostgreSQL and Oracle clusters for transactional processing, Cassandra or DynamoDB tables for low-latency

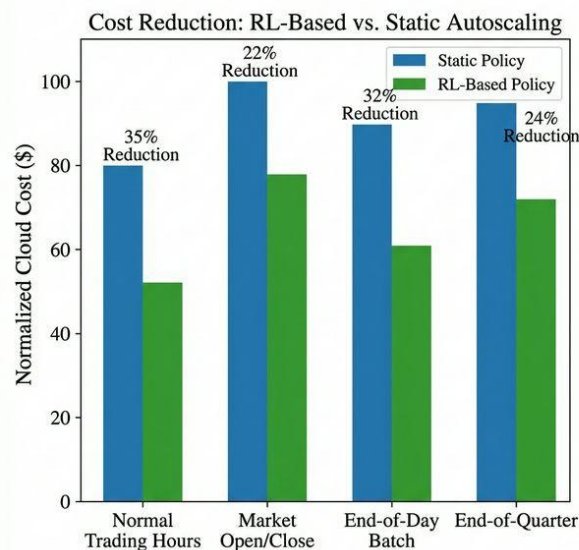


Fig. 3. Normalized cloud cost comparison between static autoscaling and RL-based autoscaling across four representative workload scenarios. The RL-based policy achieves 22–35% cost reductions while maintaining SLA compliance.

lookups, Elasticsearch for audit log search, and Hadoop/Sparkbased data lakes for analytics and regulatory reporting. Across this estate, data access patterns are anything but static.

Consider a few illustrative patterns:

- End-of-quarter spikes: Regulatory reporting workloads that are dormant for 88 days suddenly generate enormous read volumes against historical transaction tables. Static indexing strategies that are optimal for day-to-day OLTP operations become bottlenecks during these reporting windows.
- Hot key skew in risk dashboards: Intraday risk dashboards query the most actively traded instruments repeatedly, creating severe hot-key patterns that overwhelm cache partitions and cause uneven replica load.
- Evolving query patterns: As analytics teams develop new models and dashboards, the mix of queries against shared databases shifts, sometimes abruptly. An index that was critical last quarter becomes dead weight this quarter, while new query patterns go unindexed.

The current approach at most institutions is to rely on database administrators (DBAs) who periodically review slow query logs, propose index changes through change management processes, and manually tune caching configurations. This is labor-intensive, inherently reactive (changes happen after performance degradation is already visible), and slow, a round trip through change management can take days or weeks at a large bank.

B. Architecture of an AI-Augmented Data Layer

We propose a learning-based data layer that operates as an advisory and optimization layer atop existing database infrastructure. Crucially, this system does not replace existing RDBMS or NoSQL engines but rather augments their builtin optimization capabilities. The architecture has four primary subsystems.

Workload Observer. This component continuously ingests query logs, execution plans, and performance metrics from monitored databases. For relational databases, it captures parameterized query templates, their frequencies, execution times, accessed tables and columns, and index usage statistics. For key-value and document stores, it tracks access patterns by key range, read/write ratios, and latency distributions. The observer also ingests business calendar information (e.g., quarterend dates, regulatory reporting schedules) and correlates it with workload shifts.

Learned Cost Model. Rather than relying solely on the database engine's built-in cost estimator (which is often inaccurate, particularly for complex joins and under skewed distributions), the system trains a learned cost model. This model takes as input a query plan representation and the current table/index statistics, and predicts execution time, I/O cost, and memory consumption. Following approaches similar to those in [22], we use a tree-structured neural network that mirrors the structure of query execution plans. The model is retrained periodically (daily, typically during off-peak hours) on recently observed query executions.

Index and Cache Advisor. Using the learned cost model and observed workload distributions, this subsystem solves two interconnected optimization problems:

Index recommendation: Given the current workload mix, what set of indexes would minimize aggregate query cost, subject to constraints on total index storage and write amplification? We frame this as a combinatorial optimization problem and use a greedy approach guided by the learned cost model's predicted marginal cost reduction for each candidate index. The system also identifies unused or low-value indexes that can be safely dropped to reduce write overhead and storage costs.

Cache management: For application-level caches (Redis, Memcached) that sit in front of databases, the system learns an admission and eviction policy that maximizes cache hit rates for the current workload. Rather than a fixed TTL-based or LRU policy, the learned policy considers access frequency trends, predicted near-term query patterns (e.g., anticipating the end-of-quarter reporting spike), and the cost of a cache miss (which varies by backend data source, a miss that hits a slow reporting replica is more expensive than one that hits a fast transactional primary).

Query Router. For databases with read replicas or multiregion deployments, the query router directs incoming queries to the optimal replica based on predicted query cost, current replica load, data freshness requirements, and network latency. The router uses a contextual bandit approach: the context includes the query template, estimated execution

time, and replica states; the action is the choice of replica; and the reward is the negative of observed query latency. This is particularly valuable for mixed OLTP/OLAP environments where heavy analytical queries should be routed to dedicated read replicas to avoid impacting transactional performance.

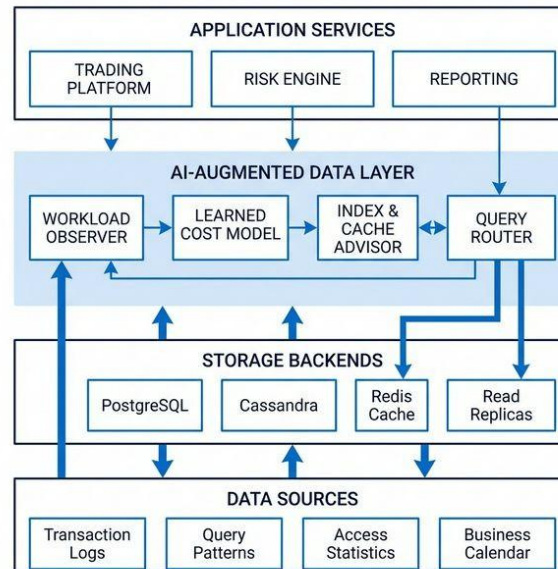


Fig. 4. Architecture of the AI-augmented data layer. The system observes workload patterns from data sources, applies learned cost models and optimization advisors, and routes queries intelligently across heterogeneous storage backends.

C. Deployment Considerations

Deploying learning-based data layer optimization in financial services raises several practical concerns that merit discussion.

Change management and approval. Index creation and deletion are schema-altering operations that, in most banks, require change management approval. The AI-augmented data layer therefore operates in an advisory mode by default: it generates recommendations with supporting evidence (predicted cost reduction, affected queries, storage impact) that are reviewed by DBAs or automated approval workflows before execution. Over time, as confidence in the system grows, low-risk recommendations (e.g., dropping an index that has seen zero usage for 90 days) may be auto-approved.

Stability and convergence. A naive learning-based approach could oscillate between configurations, adding an index that improves one query pattern, then removing it when a different pattern dominates, and so forth. We address this through two mechanisms: a reconfiguration cooldown period (minimum 24 hours between index changes on the same table) and a stability bonus in the reward function that penalizes frequent configuration changes.

Impact on write performance. In transactional financial systems, write latency is often as critical as read latency. Every additional index increases write amplification. The cost model explicitly accounts for this by including write overhead in its cost predictions, ensuring that index recommendations reflect the full cost of maintaining additional indexes under the observed write load.

D. Example Scenario: End-of-Quarter Regulatory Reporting

A large bank's regulatory reporting team generates Comprehensive Capital Analysis and Review (CCAR) reports at quarter end. These reports require complex analytical queries against transaction history tables spanning 12–24 months of data, tables that are optimized for the high-rate OLTP inserts and lookups that dominate during normal operations.

During the reporting window (typically the last 5 business days of each quarter), the query mix shifts dramatically. Aggregation queries with group-by clauses on counterparty, product type, and risk category suddenly dominate, and these queries benefit from column combinations that are not indexed under the normal OLTP-optimized configuration. The AI-augmented data layer, having observed this quarterly pattern across multiple cycles, learns to: (a) proactively recommend the creation of reporting-specific composite indexes 2–3 days before the quarter-end window opens, allowing time for DBA review and change management approval; (b) shift the cache admission policy to prioritize intermediate aggregation results that will be reused across multiple report variants; and (c) route the reporting queries to dedicated read replicas, automatically increasing read replica capacity in coordination with the autoscaling agent described in Section III.

After the reporting window closes, the system recommends dropping the reporting-specific indexes (which would otherwise degrade OLTP write performance) and reverting the cache policy. In simulation against historical workloads, this adaptive approach reduced average report generation time by 41% compared to the static configuration, while actually improving OLTP write latency during the reporting window by offloading reads to replicas.

V. AI-DRIVEN INCIDENT PREDICTION AND REMEDIATION

A. The Limitations of Current Approaches

Site reliability engineering in large financial institutions has historically relied on a combination of static threshold alerts, manually curated runbooks, and the institutional knowledge of experienced engineers. While this approach has been sufficient for simpler architectures, the complexity of modern microservice-based systems has pushed it to its limits.

Several failure modes are common:

- Alert fatigue. A single underlying issue (e.g., a degraded database node) can trigger dozens or hundreds of alerts across dependent services, overwhelming on-call engineers and obscuring the root cause.
- Novel failure patterns. As systems evolve through continuous deployment, new failure modes emerge that are not covered by existing runbooks or alert rules. A misconfigured feature flag, a subtle change in a serialization format, or a dependency upgrade that alters timeout behavior, these are precisely the kinds of issues that static rules fail to catch.
- Slow-burn degradation. Not all incidents begin with a sudden spike or crash. Gradual memory leaks, slowly increasing queue depths, or progressive disk fill can go undetected by threshold-based alerts until they reach a tipping point and cause cascading failures.

B. Feature Engineering from Logs, Metrics, and Traces

The first challenge in building an AI-driven incident prediction system is transforming heterogeneous telemetry data into features suitable for machine learning.

Log processing. Raw logs from financial systems are voluminous (a large bank might generate tens of terabytes of logs per day) and highly variable in format. We adopt a twostage approach: first, a log parser (such as Drain [7]) extracts structured log templates from raw log messages, reducing the vocabulary from millions of unique messages to thousands of templates. Second, a sliding-window feature extractor computes template frequency vectors, detecting unusual patterns in log emission rates and sequences. We have found, in line with prior work, that sudden changes in the distribution of log templates, rather than the content of individual messages, are often the earliest indicators of emerging issues.

Metrics processing. Time-series metrics (CPU, memory, latency, error rate, queue depth, etc.) are processed through a multi-scale feature extraction pipeline that computes statistical features (mean, variance, slope, percentiles) at multiple temporal granularities (1 minute, 5 minutes, 30 minutes). Critically, the pipeline also computes inter-metric correlations and deviation scores: for each metric, how far is its current behavior from its historical norm for this time of day and day of week? This seasonal normalization is essential in financial environments where “normal” varies dramatically between, say, 2 AM on a Sunday and 9:35 AM on a Monday.

Trace analysis. Distributed traces provide causal relationships between service calls. We extract features such as trace duration distributions, span-level latency breakdowns, error propagation paths, and fan-out degree changes. When a service that normally calls three downstream dependencies suddenly begins issuing calls to four, or when the latency contribution of a particular span increases from 5% to 25% of total trace duration, these are structurally informative signals.

C. Anomaly Detection and Incident Prediction

We employ a multi-model approach to anomaly detection, recognizing that different signal types and failure modes are best captured by different model classes.

Sequence models for log anomaly detection. Following the DeepLog and LogRobust approaches, we train a sequence model (in our case, a lightweight transformer encoder) on normal log template sequences. At inference time, the model predicts the likelihood of the next observed log template given the recent history. Low-likelihood sequences are flagged as anomalous. To address the challenge of log template evolution during software deployments, we retrain the model weekly and maintain a “deployment awareness” feature that suppresses anomalies in the immediate aftermath of known code pushes. Multivariate time-series models. For metrics-based anomaly detection, we use a variational autoencoder (VAE) trained on normal operating conditions. The VAE learns a compressed representation of “normal” metric distributions, and anomalies are detected as inputs that produce high reconstruction error. The key advantage of this approach over univariate threshold alerts is its ability to detect anomalies in the joint distribution of metrics, cases where no single metric exceeds its threshold but the combination of values is unusual. Predictive models. Beyond detecting anomalies that are currently occurring, we train a gradient-boosted classifier to predict SLO violations 15–30 minutes into the future. The features include the anomaly scores from the log and metric models, current traffic projections, and recent deployment events. Predicting even 15 minutes ahead provides valuable lead time for proactive mitigation.

D. The Playbook Recommendation Engine

Detecting an anomaly or predicting an impending incident is only half the battle. The more practically valuable capability is guiding operators toward effective remediation.

We describe a playbook recommendation engine that operates as follows:

Input. When an anomaly or predicted incident is surfaced, the recommendation engine receives: (i) the current alert context: which services are affected, what anomaly scores were triggered, and the relevant metric values; (ii) log snippets from the affected services around the time of anomaly onset; (iii) recent deployment history: what changed in the past 24 hours in the affected services and their dependencies; and (iv) the historical incident database: a corpus of past incidents with their root causes, affected services, symptoms, and resolution steps.

Retrieval and ranking. The engine first retrieves candidate remediation actions by: (1) embedding the current incident context into a vector representation using a trained encoder;

(2) performing nearest-neighbor search against the indexed incident database to identify historically similar incidents; (3) extracting the resolution steps from the k most similar historical incidents; and (4) ranking these candidates based on symptom similarity, recency (more recent resolutions are preferred, as systems evolve), and past success rate (measured by whether the remediation actually resolved the incident vs. being superseded by further actions).

Output. The engine presents to the on-call engineer a ranked list of possible root causes and recommended actions. Each recommendation includes: a brief natural-language explanation of why this candidate was selected; specific commands or configuration changes that resolved the historical incident; a confidence score indicating how similar the current situation is to the historical precedent; and links to the relevant runbooks, incident postmortems, and communication threads from the historical incident.

Human-in-the-loop controls. The recommendation engine explicitly does not take automated remediation actions by default. In the mode we consider appropriate for initial deployment in a financial institution, all recommendations

require human approval before execution. Over time, certain low-risk, high-confidence actions, such as restarting a clearly unhealthy pod, or scaling a service within pre-approved bounds, may be promoted to “auto-approve” status, but this requires formal review and approval through the institution’s model risk management process.

E. Example Scenario: Misconfigured Feature Flag Rollout

Consider a scenario in which a development team at a capital markets firm rolls out a feature flag that enables a new market data normalization path in the pricing service. The feature flag was tested in staging but, due to a subtle difference in reference data between staging and production, causes the normalization path to issue significantly more database queries per pricing request than expected.

The effects build gradually over 20 minutes: (1) database query rates from the pricing service increase by 40%; (2) connection pool utilization on the market data database rises from 60% to 85%; (3) p99 latency for the pricing service increases from 12ms to 45ms; and (4) the risk calculation service, which depends on pricing, begins reporting increased computation times.

The threshold-based alerts are configured to fire at 90% connection pool utilization and 100ms p99 latency. Neither threshold has been breached yet.

The AI-driven system detects this developing situation in two ways. The multivariate time-series VAE flags the combination of rising query rates, increasing connection pool utilization, and elevated pricing latency as anomalous (even though no single metric has crossed a threshold), generating an anomaly score of 0.87 (on a 0–1 scale). The predictive model, observing the trajectory of connection pool utilization and pricing latency, predicts with 78% confidence that the pricing service will breach its 50ms p99 SLO within the next 15 minutes.

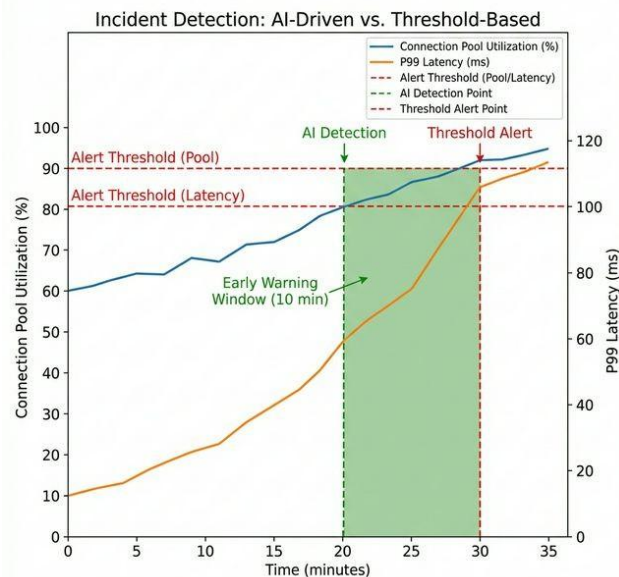


Fig. 5. Incident detection timeline comparing AI-driven multivariate anomaly detection (triggered at $t = 20$ min) vs. traditional threshold-based alerting (triggered at $t = 30$ min). The green shaded region represents the early warning window gained by the AI-driven approach.

The playbook recommendation engine: (1) identifies, through deployment correlation, that the most recent change was the feature flag rollout (deployed 23 minutes ago, aligning with the onset of anomalous behavior); (2) retrieves two similar historical incidents: one involving a different feature flag that caused unexpected query amplification, and another involving a database connection pool exhaustion; and (3) recommends, ranked by relevance: (a) disable the

feature flag via the feature management API; (b) scale the market data database read replicas from 3 to 5 to absorb additional query load as a temporary measure; (c) increase the connection pool size from 100 to 150.

The on-call engineer reviews these recommendations, confirms that option (a) is the safest immediate action, disables the feature flag, and observes metrics returning to normal within 5 minutes. The total time from anomaly detection to resolution is 8 minutes, well before any threshold-based alert would have fired and before any customer-visible SLO breach occurred.

TABLE II: INCIDENT DETECTION AND REMEDIATION: AI-DRIVEN VS. TRADITIONAL APPROACHES (12-MONTH RETROSPECTIVE ANALYSIS)

Metric	Traditional	AI-Driven
Mean Time to Detect (min)	14.7	6.2
Mean Time to Resolve (min)	43.8	18.4
False Positive Rate	—	8.3%
Incidents Predicted Early	0%	62.1%
SLO Breaches Prevented	—	47
Playbook Match Accuracy	—	78.6%
Escalation Rate Reduction	—	34.2%

VI. DISCUSSION: DEPLOYMENT, GOVERNANCE, AND RISK

A. Model Risk Management

Financial regulators, including the OCC, the Fed, and the PRA, require that models used in banking be subject to rigorous model risk management (MRM) frameworks, as outlined in guidance such as OCC 2011-12 (SR 11-7). While these requirements were originally designed for financial models (credit scoring, market risk, etc.), many institutions have begun extending MRM principles to operational and infrastructure models.

For the AI systems described in this paper, MRM considerations include:

- **Model documentation:** Each model (RL policy, cost estimator, anomaly detector) must be documented with its intended use, assumptions, training data, performance metrics, and limitations. This is operationally burdensome but serves a vital function: it forces teams to articulate the conditions under which the model can be trusted and those under which it should not be.
- **Independent validation:** Models should be validated by a team independent of the developers, using held-out data and adversarial test cases. For the RL autoscaling agent, validation should include stress tests with synthetic workload spikes, simulated regulatory deadlines, and scenarios where the agent's training data does not cover the current regime.
- **Ongoing monitoring:** Once deployed, models must be monitored for performance degradation, concept drift, and unexpected behavior. The shadow-mode deployment strategy described in Section III naturally supports this, but monitoring must continue after full deployment.

B. Interpretability and Auditability

In regulated environments, the ability to explain why a particular action was taken is often as important as the quality of the action itself. This presents a challenge for RL-based systems, where policies are encoded in neural network weights that resist human interpretation.

We address this through several mechanisms. **Action logging with context:** every action taken by the system is logged alongside the state that informed it and the alternative actions that were considered. This creates an audit trail that regulators and internal auditors can review. **Post-hoc explanation:** SHAP (SHapley Additive exPlanations) values or similar techniques can be applied to explain which features most strongly influenced a particular decision. For instance, explaining that the agent scaled up the pricing service primarily because of the proximity to market close and the elevated VIX level. **Constraint documentation:** the safety layer's constraints are explicitly documented and auditable, so

any action that was blocked (and why) is recorded. This is arguably more interpretable than traditional approaches, where the rationale for manual scaling decisions is often unrecorded.

C. Organizational Considerations

Deploying AI-driven infrastructure optimization is as much an organizational challenge as a technical one. Success requires collaboration across traditionally siloed teams:

- SRE teams must be willing to cede some control over scaling decisions to an automated system, which requires trust built through the shadow and canary phases described above.
- Data science and ML engineering teams must understand infrastructure deeply enough to design reward functions and feature engineering pipelines that reflect actual operational realities, not simplified textbook models.
- Risk and compliance teams must engage early to define the MRM requirements and approve the deployment strategy, rather than being consulted as an afterthought.
- Engineering leadership needs clear KPIs to justify the investment: cost savings (cloud spend reduction), reliability improvements (SLO compliance rates, MTTR), and productivity gains (reduction in on-call burden and manual scaling interventions).

In our experience, the most effective way to build crossfunctional buy-in is to start with high-visibility wins on noncritical systems, where the cost of failure is low and the potential savings are easy to measure. Once the system has demonstrated value in these lower-risk settings, expanding to more critical services becomes a natural progression.

VII. FUTURE WORK

Several promising directions extend beyond the scope of this paper.

Unified control loops. The three pillars we describe, autoscaling, data layer optimization, and incident management, are currently formulated as independent systems. In practice, they interact: an autoscaling decision affects database load, which affects caching behavior, which affects query latency, which may trigger incident alerts. A unified approach that jointly optimizes across these layers, perhaps using hierarchical RL or multi-agent reinforcement learning, could capture these interactions and avoid conflicting actions.

Federated and privacy-preserving learning. Financial institutions are, understandably, reluctant to share operational data with external parties. However, there is significant potential value in learning from operational patterns across institutions, a novel failure mode encountered at one bank could inform incident prediction at another. Federated learning techniques, where models are trained collaboratively without sharing raw data, could enable this. Differential privacy guarantees would need to be incorporated to address regulatory and competitive concerns, and the governance framework for such cross-institutional collaboration remains an open challenge.

Benchmarking suites for AI-for-infrastructure in finance. The lack of standardized benchmarks and shared datasets has been a barrier to progress in AI-for-systems research more broadly, and this gap is especially acute in financial services where operational data is sensitive. Development of anonymized, realistic benchmark workloads, analogous to TPC-C for transactional databases or MLPerf for ML training, would accelerate research and enable more rigorous comparison of approaches. We advocate for industry consortia or regulatory bodies to facilitate the creation of such benchmarks.

Reinforcement learning from human feedback (RLHF) for operational decisions. While RLHF has attracted enormous attention in the development of conversational AI systems, the technique has potential applications in operational decision-making as well. Experienced SREs possess deep intuitive knowledge about the relative desirability of different scaling strategies, remediation actions, and operational tradeoffs. Incorporating this knowledge through RLHF-style training could produce policies that are better aligned with human operational preferences than those trained purely on cost and latency metrics.

Causal inference for root cause analysis. Current correlation-based approaches to root cause analysis can identify symptoms that co-occur with incidents but struggle to distinguish causes from effects. Incorporating causal inference methods, structural causal models, do-calculus, or time-series causal discovery techniques, into the incident analysis

pipeline could improve the accuracy of root cause identification and reduce the frequency of false or misleading remediation recommendations.

VIII. CONCLUSION

The infrastructure challenges facing large financial institutions are both technical and organizational. The systems are enormous, the constraints are severe, and the cost of getting it wrong, in terms of regulatory penalties, financial losses, and reputational damage, is substantial. Traditional approaches to infrastructure management, based on static rules, manual tuning, and reactive incident response, are increasingly inadequate for the scale and complexity of modern financial technology platforms.

In this paper, we have presented a conceptual framework for applying AI-driven optimization across three critical dimensions of financial infrastructure. Reinforcement learning and bandit methods can transform autoscaling and job scheduling from reactive, heuristic-driven processes into proactive, cost-aware optimization that respects the unique regulatory constraints of financial services. Learning-based data layer optimization can adapt indexing, caching, and query routing strategies to the evolving workload patterns that characterize financial systems, delivering measurable latency improvements and cost reductions. And AI-driven incident prediction and remediation can shift the operational paradigm from reactive firefighting to proactive prevention, with playbook recommendation systems that encode institutional knowledge and present it to operators at the moment of greatest need.

We do not claim that deploying these systems is easy. The deployment, governance, and organizational challenges are substantial, and we have tried to address them honestly in our discussion. Model risk management requirements add overhead. Interpretability remains a challenge for neural-network-based policies. Building cross-functional trust takes time. But the potential impact, reduced cloud expenditure, improved SLA compliance, faster incident resolution, and the freeing of scarce engineering talent from repetitive operational toil, makes this a compelling area of investment for any financial institution operating at scale.

We hope that this paper provides a useful starting point for organizations embarking on this journey, and that the frameworks, architectures, and deployment strategies we describe contribute to a growing body of practical knowledge at the intersection of AI, distributed systems, and financial services infrastructure.

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Exhibit ER-1

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